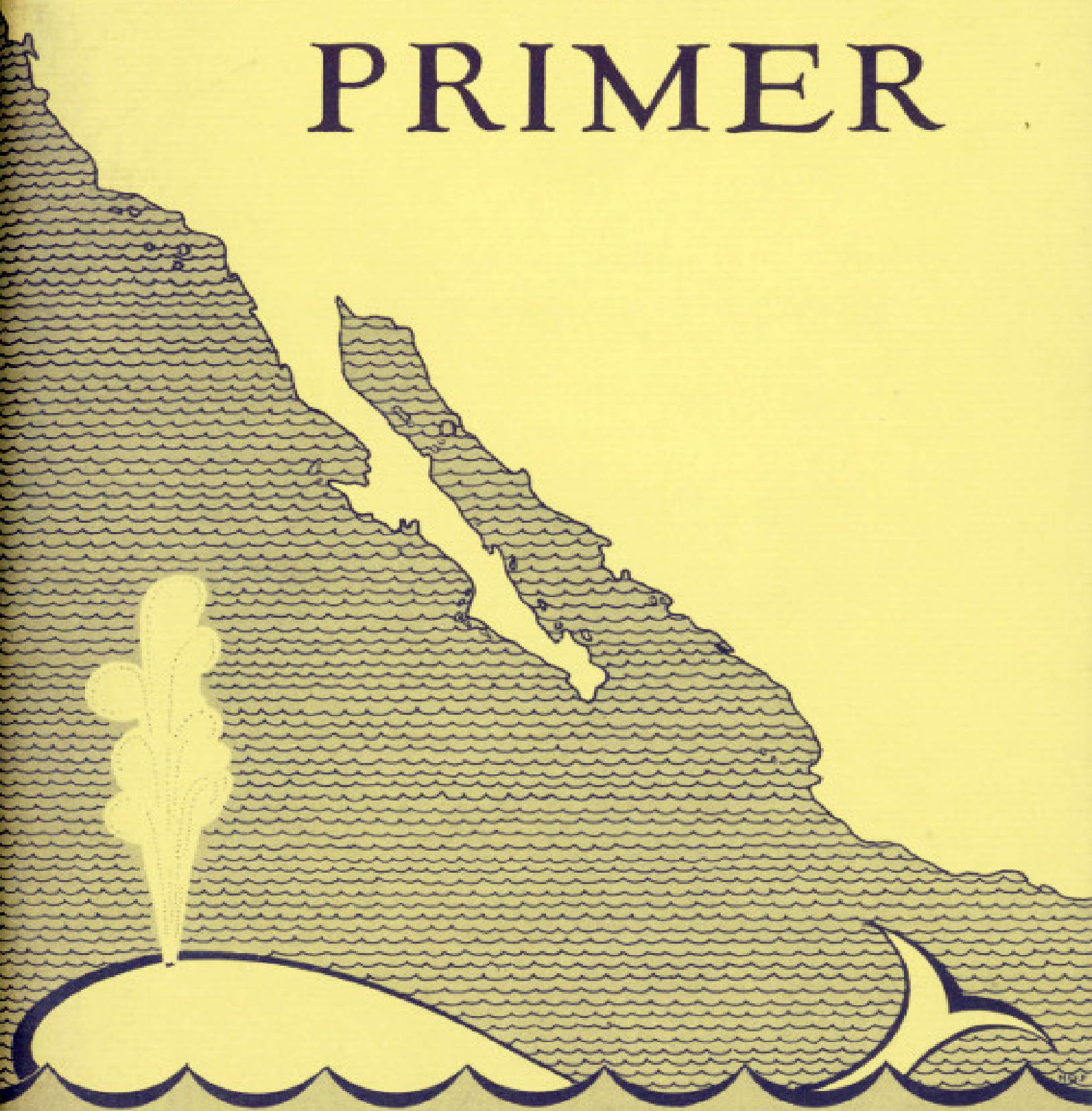
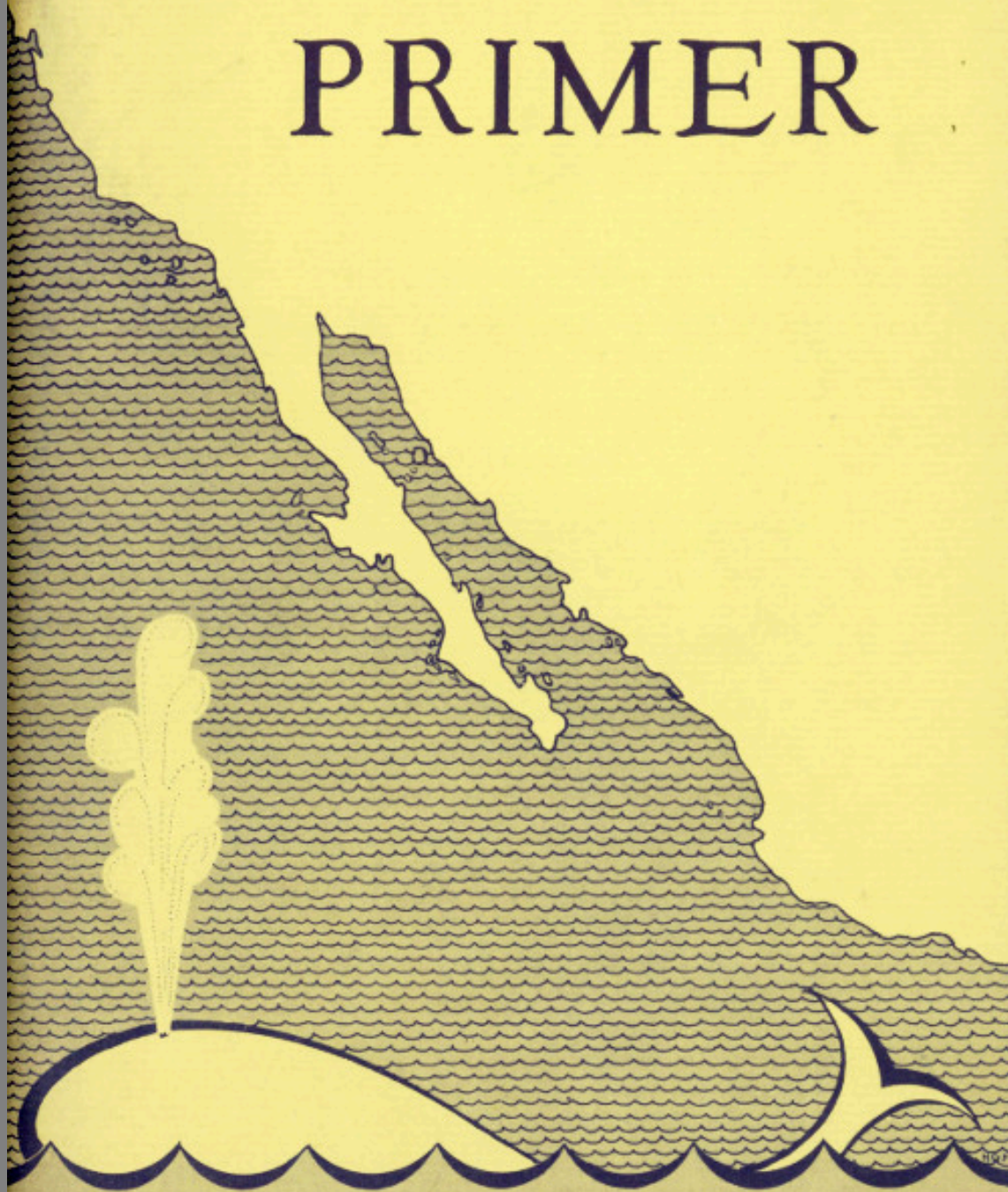


WHALE PRIMER



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PRIMER, WITH SPECIAL ATTENTION TO THE CALIFORNIA GRAY
WHALE ***



WHALE PRIMER

With Special Attention to
THE CALIFORNIA GRAY WHALE

by
Theodore J. Walker

Published by
the
Cabrillo Historical
Association
1962



Produced in cooperation
with the National
Park Service

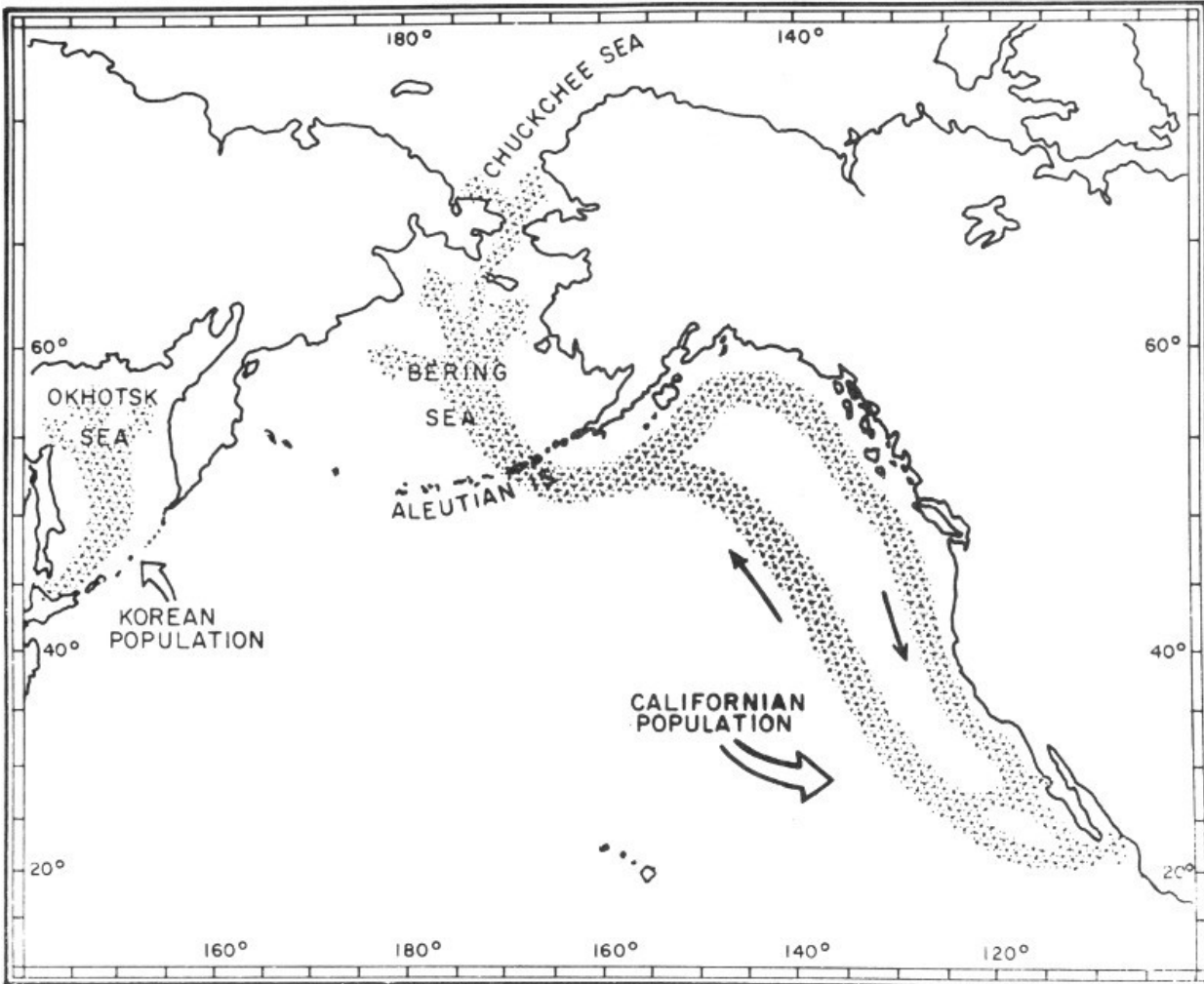
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1962

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Migration routes of the California gray whale. The Korean herd may now be extinct.

Introduction

The WHALE PRIMER provides a brief introduction to one of nature's most interesting creations, the whale. The principal star of the handbook is the California gray whale which in recent years has become a major tourist attraction in southern California. Notwithstanding the extreme interest, no concise interpretation of the migration has been prepared. Although there is a tremendous number of technical and popular writings about whales, there is still great mystery about them. Whales carry on practically their entire lives below the surface of the sea out of reach of man, so that most of our knowledge has been pieced together from the study of the bodies of slaughtered whales. The literature abounds in partial truths, misinterpretations and technicalities which confuse even the specialists. Many of the sources of information require translation.

Furthermore, many of these papers were published in journals of limited distribution. Others are long since out of print, and much of the primary historic records can be found by examination of records which exist only in one particular library. In the preparation of this manuscript, hundreds of books and over 4,000 papers were catalogued, of which the most important were available, and examined. The author was particularly fortunate to have Japanese and Russian friends who gave gladly of their time to insure coverage of these important papers.

The author deliberately made an extreme condensation of the facts in order to prevent the reader from being overwhelmed by details that merely obscure the broad picture. It is hoped that the reader will gain an awareness of the extreme mastery by whales of the marine environment. Other basic concepts of biology, which are clearly illustrated by the natural history of whales, are developed.

The intense interest and pleasure which the sight of the migrating whale creates clearly overshadows the brief monetary benefit that the whaling industry might gain from slaughtering it. We hope that you will be

stimulated to join forces with those of us who feel that man should preserve those forms of life which add so much interest, beauty, and knowledge to man's awareness.

Two General Migration Routes

The migration of the California gray whale is one of the most remarkable natural history events in the world today. The majority of these whales journey southward just off the shore of southern California and Lower California during January and February. Although a few early migrants may pass San Diego early in December, they are not abundant until Christmas. An occasional straggler can be sighted in March.

Only 20 years ago this species was so rare, that little hope was held that it could ever recover. Today the species appears out of danger, thanks to international cooperation among the whaling nations which stopped the slaughter of this truly unique whale. Now it is not at all unusual to see between 50 and 75 whales a day during the peak of the migration.

One of the finest locations for viewing this migration is the Cabrillo National Monument which commands an almost aerial view of the coastline. Here individual whales can be watched for at least 1 hour, as they hove into view from the north and at last recede to the southeast along the Silver Strand. The first and only public observatory for whales was established at the monument in 1951. The naturalist on duty not only keeps a plot of the whales passing by, but also helps visitors find their first whale. The observatory is one of the most popular wintertime attractions in southern California. With so many pairs of eyes on hand, it is not surprising that the count of whales is remarkably complete. There is no other marine animal which can be seen with such certainty in its natural element.



Migrating gray whales off Point Loma, Calif. *Photograph by Burky Reeves.*

4



Migrating gray whale passing San Diego, Calif. *Courtesy Scripps Institute of Oceanography.*

Migrating animals have always fascinated man who considered them harbingers of the seasons. Man continues to puzzle over the mysteries

5

of how these animals are able to navigate so precisely and how they are able to maintain such timetables. Whereas other migrating animals pass broadly through an area, the California gray whales, at least on the final part of the route, are passing along just outside the surf zone, virtually single file! It is hard to realize that 3 months earlier these whales started off from their summer quarters in the Arctic Ocean and the Bering Sea, as well as along the shore of Siberia and Kamchatka en route to their winter quarters in the lagoons and harbors along the outer coast of Lower California. Between these two areas lie 6,000 miles of seemingly trackless ocean. With the advent of spring the whales must be on their way back again to their summer grounds.

Although all the large whales make such extensive travels, except the bowhead, only the gray whale spends so much time in sight of land. The other species are truly oceanic at all times, and never seem abundant because of the vastness of the oceans. Like the gray whale they congregate in polar seas during the summer months, moving into temperate and subtropical waters for the winter months. To this day, much of the migration route is unknown. Perhaps some day a scientist will attempt to trail a group of whales along the entire route. The tendency of the gray whale to hug the coast is manifest only within 600 miles of the destination. This may be a precautionary routing which prevents the whales from making their landfall south of the lagoons. Such an error in navigation would not only prolong the migration, but leave the whales on the horns of a dilemma—to swim on south or turn back?

Whale Waters—Summer and Winter

One cannot help but be impressed with the remarkable utilization of time by the whales whose lives seem to be divided into two principal seasons, a summer feeding period and a winter period of reproduction. Each of these major activities is preceded by a tremendously long migration. Nearly half of every year must be devoted to this activity. Considering the extreme length of the migration, whales cannot wander aimlessly or carelessly. Whales which summer in the Antarctic continue to do so as do the whales in the Arctic waters, and only rarely does one pass through the wide belt of equatorial water to venture into the other hemisphere.

Once on the summer grounds the whales occupy themselves with feeding almost continuously during the long polar day. Even though the food is patchy, the whales seem to find it quickly, spending a minimum of time in search. By the onset of autumn, they are fat, and all the babies are weaned.

It is uncanny that the various species all manifest the instinctive 6 reaction to vacate this region at the proper time, thus avoiding almost certain death by the freezing of the sea's surface. Again the seeming miracle of aptness is evident, for the whales swim unerringly out of the dangerous areas toward warmer and calmer seas. Because of the extremely wide band of winter storms, whales must move at least below 30° latitude to be clear of the areas of stormy seas. Migration stops as soon as they are sufficiently clear of these. Whales then undertake the other essential link in the chain of life, reproduction. By spring the babies are strong enough and fat enough to accompany their mothers.

Feeding—Filter Whales

Whales do not feed extensively while migrating. For the most part there is not time enough, nor is the food plentiful enough to make it worth the effort. However, in the polar seas the whale's food is plentiful enough to discolor the water. On close examination, the discoloration proves to be caused by thousands of tiny shrimps which are very slender and less than one-half inch in length. These creatures congregate in swarms near the surface to feed on microscopic plants known as diatoms. The whales need only swim back and forth through these cloudlike aggregations to fill their mouths quickly with water and shrimp. With each mouthful, the water is expelled between the jaws through a mat of fibers which hangs down from the upper jaw. The shrimps, which are retained on the mat, fall down onto the tongue and are swallowed. The work of pushing out several tons of water with each feeding is done by the tremendous tongue.

The fiber mats are the frayed inner side of enormous hornlike plates which grow down from the palate. The main body of each plate is placed edgewise to the outgoing water so that many plates are required to complete the mat which runs from the tip of the jaw to the corner of the mouth. These plates vary in size and stiffness from species to species. Some of the plates from the mouth of the bowhead whale are 12-14 feet in length, whereas in the finback whale the plates are 2-4 feet. There can be over 200 plates per side in the filtering structure. The frayed inner edge is constantly breaking 7 off and the plates keep growing and fraying to provide the necessary thickness for the mat. Technically, these plates are called *baleen*. They were called whalebone by the whalers, and that is the name which is still used in commerce. The plates have no relationship to bones, nor could they be mistaken for them. The whalebone was assiduously collected and sold to be made into a variety of objects such as umbrella stays, corset stays, buggy whips and other articles which today are made of steel or plastic. There was a great demand for the product and a bowhead whale produced over a ton and one-half of whalebone valued back in the 17th century at over 400 English pounds, equivalent to about \$10,000 today.

It has been observed that the coarseness and thickness of the baleen is suited to the size of the food which is filtered. For example, the rorqual or Sei whale, which feeds on tiny shrimp species, has a filtering surface which resembles fine wool. The blue whale, which feeds on the largest of shrimps and on fish, has the coarsest filter. Generally, fish occur only in the diets of the blue, finback and other rorqual which swim fast enough to engulf them. Here, the fish are weak-swimming, schooling fishes. The gray whale, unlike the other filter feeders, feeds on bottom-frequenting crustaceans known technically as amphipods. These organisms, occurring principally in shallow water, keep the gray whales close to the shores of Siberia and Kamchatka.

In order for a whale to be able to exist on a 3 to 4 month feeding period, it must have not only ample food, but time in which to collect it. Although the whales may not all be far enough north to have a 24-hour day, there is enough twilight to let them feed the clock around. In order to take advantage of the prodigious amounts of food available, whales have a huge four-chambered stomach. It is not at all uncommon to find 5 to 10 wheelbarrow loads of shrimp in the stomach. No one yet knows how fast the food passes through the alimentary tract.

The shrimps on which the whales thrive are not uniformly concentrated, but occur at special places where oceanographic processes have enriched the surface waters with cold, nutrient-rich, subsurface water. The principal oceanographic action which enriches the water at the surface is called upwelling. It can be induced by a number of physical conditions. One of the best places to look for upwelling is along the edge of the polar icecaps and along the junction of strong currents. Upwelling is one of the ways nature refertilizes the surface waters which are otherwise deficient in nutrients. Whenever the ocean surface is fertilized, the microscopic plants begin to grow and multiply, discoloring the water to a brownish or reddish color. 8 If the subsurface waters continue to be pumped to the surface, the growth continues and a rich ocean pasture results. The animals which begin to crop this are the shrimps, and when they have thrived and reproduced, there are enough for the whales. The other all-important factor in the fertility of the polar seas is the length of day, which provides ample time for the sun's energy to be entrapped by plants. By autumn, the days have shortened and night begins to predominate. At this time the microscopic plants stop growing and form resting stages which protect them through the long winter.

The whale has made such efficient use of its summer feeding period that it is fat enough to live the other 8 months without danger of starvation. Whales might be thought of as huge natural tankers carrying enough oil to provide for long periods of active fasting. Unlike those mammals which avoid starvation by winter hibernation, the whale is able to migrate away from inhospitable seas into warmer waters where they can then reproduce and care for the new young. The strain on the mother at this time must be considerable for her baby grows at a prodigious rate. It has been calculated that a blue whale baby grows about 10 pounds per hour, gaining a ton every 9 days.

Although there are at least nine species of filtering whales, each seems to be dependent on a different kind of shrimp which are seldom found together. This prevents the various species of whales from being in constant competition with each other. However, no one altogether understands why there is no competition. Of course, structurally a species may be more suited to feeding on one kind of food. The porosity of the filter plate, which varies, may account for this. The actual shape and size of the head varies from species to species and this may be related to improving the efficiency of feeding. Right whales have heads which are very large, permitting a large intake of water and the accommodation of a tremendous set of filters. These whales have heads which are enlarged both in width and length. In fact, the head may account for nearly one-third of the total body. The rorqual group are more streamlined and have proportionately smaller heads. To make up for this limitation, the floor of the mouth is pleated on the outside which permits the floor to balloon out like a huge scoop each time the mouth is filled.

The right whales, incidentally, were so named by the early whalers to apply to those species which were suitable for whaling. The majority of whales were not molested because they sank as they died, or they were fast-moving, wary species which could not be approached with the whaler's lance.



Infra-red photograph of gray whale reveals heat of the spout and also indicates the double source of the spout. *Photograph by T. J. Walker.*

Evolution of Whales

To appreciate fully the biology of whales, one must know that their ancestors were terrestrial mammals. It is indeed impossible to account for all the steps which were necessary for this difficult reentry of the oceans. However, the fossil records for whales are numerous and permit at least a partial reconstruction of the evolutionary steps. Whales have obtained not only complete mastery of this difficult habitat, but also they have spread out to crop a variety of marine foods. Biologists have generally dramatized the earlier conquest of land by marine organisms, leaving the more recent and perhaps more difficult reentry of the marine world by whales to go unnoticed.

Breathing Adaptations

Not only have whales become completely aquatic, but they have been able to eliminate nearly all the design features which were necessary for life on land. Only the retention of air breathing remains, and this does not seem to be much of a hardship. There has been a great improvement in the conservation of oxygen so that really long dives are possible. The subtlety of this accomplishment is only partially understood by scientists. Apparently, diving mammals are able to shut down those bodily activities which contribute little to the diving mission. These activities can go on later when oxygen is available. It is also normal to incur an oxygen debt by borrowing from stockpiles present in the tissue fluids and muscles. After a long dive a whale will idle at the surface in order to completely free the body of the excess carbon dioxide, and to pay back the oxygen debt. The greater the debt the longer the surfacing, and the greater the number of breaths which must be taken.

When a whale surfaces to breathe, the act of exhaling is called “blowing.” Whenever a whale has been submerged for a normal dive, the air in the

lungs becomes saturated with moisture from the blood. The exhalation of this spent air is accomplished very quickly by forcing the air out under pressure by the diaphragm, and the sudden expansion of the expelled air produces sufficient cooling to condense the moisture. This cloud or fog is the most conspicuous feature of a surfaced whale, particularly when the spout is 10 to 15 feet in height. Within a minute's time the fog is usually scattered and heated enough to disappear. The duration of the spout depends 11 principally on the temperature of the surrounding air, the amount of moisture condensed from the breath and the local surface wind. At the higher latitudes air temperatures are low enough that the spout may persist for several minutes. It is possible to recognize some of the whale species by the form and size of the spout.

Inhalation is accomplished very quickly. The breathing act is generally both visible and audible. The release of air produces a very loud "whoosh" which can be heard for quite a distance on a quiet day. The nostrils are called blowholes. In order to facilitate breathing, they have been moved from the tip of the snout to the top of the head (with the exception of the sperm whale), to prevent waves from flooding the lungs. During diving, the pressure of the water operates on the nostril in such a way as to close the nostril from the outside so that regardless of depth there can be no leak. The natural buoyancy of the animal exposes enough of the head to keep the nostrils clear of the waves.

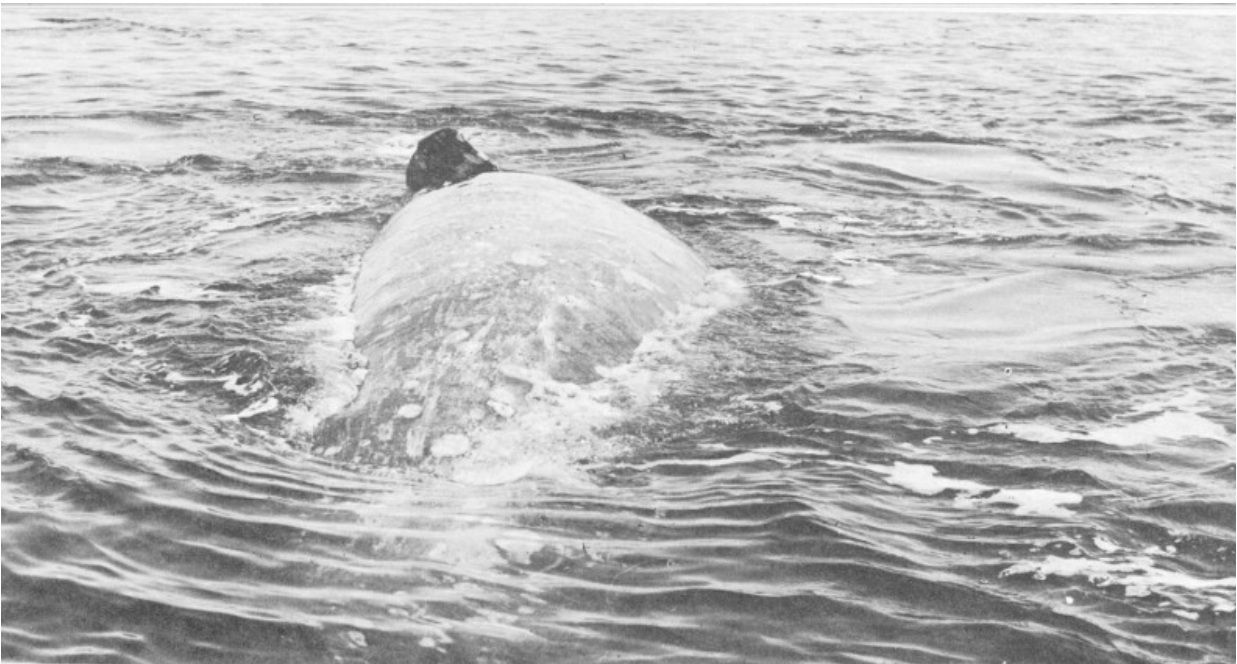
The nostrils communicate directly to the lungs rather than share a portion of the throat as is customary in other air-breathing vertebrates. This means that the whale's mouth and throat can be full of water without danger of flooding the lungs, and that it is unnecessary to empty such a spacious cavern prior to breathing. Furthermore, it is unlikely that a whale could keep his mouth closed enough to prevent flooding through the baleen because there is no upper lip over this device.

Although a whale can be sighted by the telltale spout, a frightened whale may elude detection by exhaling just before surfacing, so that nothing more than a foamy patch is produced. Under these situations the whale does not expose the usual amount of buoyant head, but only the nostrils. A disturbed whale can dive, and then surface a mile or two away, or it may not move at all, preferring to hide on the bottom or among rocky reefs or in the kelp. The

California gray whale was judged by whalers to be the most wary and elusive of them all.

Swimming Adaptations

The most essential features needed for the successful invasion of the marine habitat were those necessary for efficient propulsion. Fish, eons before, had solved the hydrodynamic equations necessary for movement through such a resistive medium. This solution required a streamlined form with a tail for propulsion, placed at the very end of the body. Extra fins were employed for maneuvering and for balancing. Whales, too, have reached the same solution, and man, when he finally develops sufficient highspeed submarines, will employ the same solution, namely streamlining. As a consequence, all whales look alike, differing principally in the degree of streamlining, color or size. In a whale's streamlined body there can be no sharp discontinuities to accommodate the head, the neck, the trunk, and lastly the tail. Instead these features must grade imperceptibly one into the other. The only allowable discontinuity is the end of the tail which is expanded into fanshaped lobes to function like a propeller.



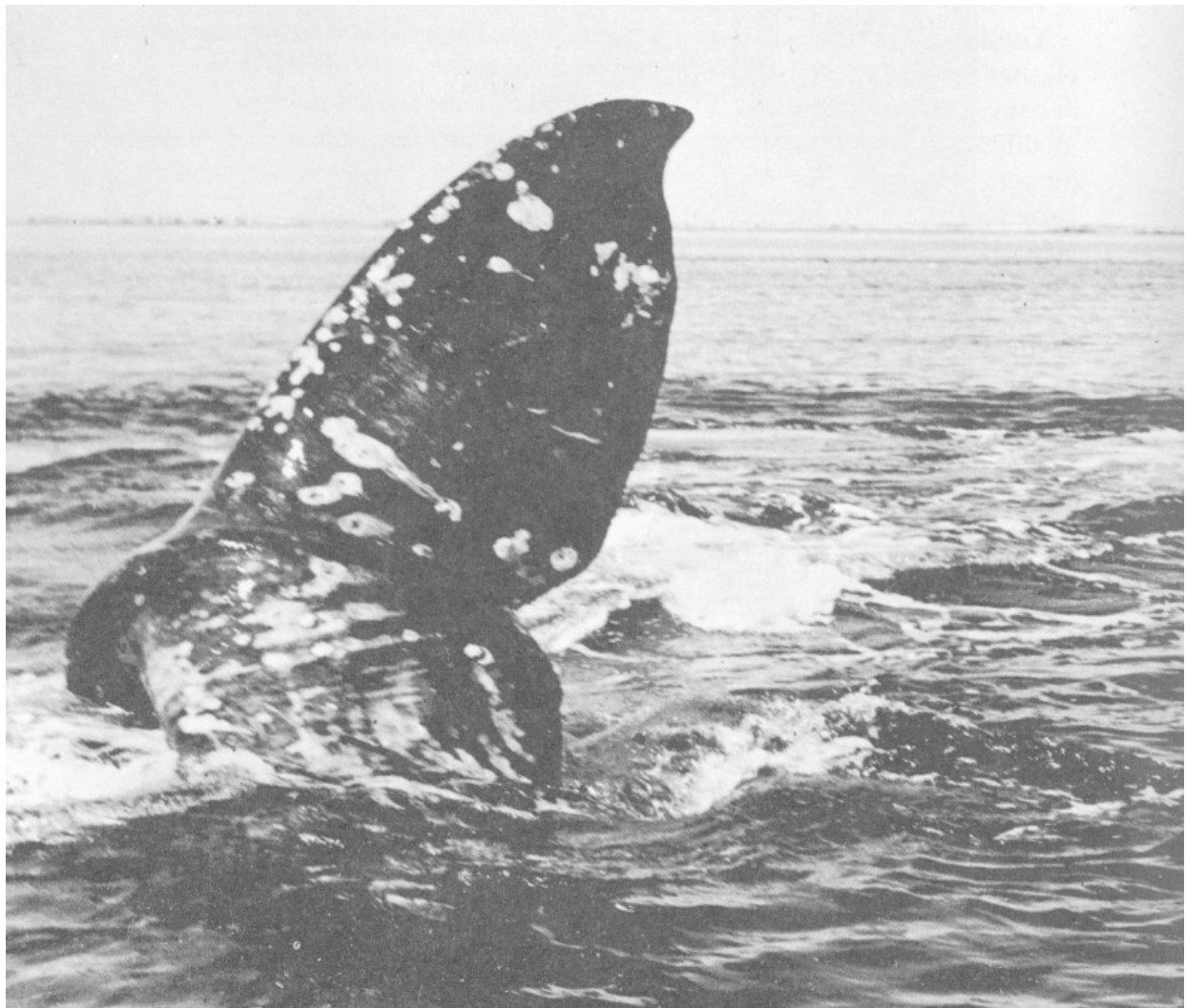
Gray whale rolled over on side during courtship. Note outline under water of tail and tailflukes.
Courtesy Scripps Institute of Oceanography.

These features, called tail flukes, are driven up and down in contrast to the tail of a fish which is driven sidewise. Whales have long banks of muscle along either side of the backbone which attach to the tail flukes by means of tendons. This makes it unnecessary to disturb the streamlined form by bending the hind part of the trunk as is necessary when fish drive with their tails. It also makes it possible to devote a great deal more muscle to the task. The power developed by these muscles is prodigious, capable of driving a 100-ton body through the water at speeds up to 20 knots. Wounded whales can smash a 20-foot whaleboat to bits with a single slap of the tail. 13

The hind limbs which were useful on land have been eliminated and all that persists are vestigial bones or cartilages which are buried deep below the surface of the body. The forelimbs have undergone reduction and modification into flippers which assist in the turning and diving. The flippers are useful in other ways, providing a platform on which the baby may stay when danger threatens. They are also useful during courtship and mating, but not for combat. The toothless whales do not have too much to fight with. They may strike an adversary with the powerful tail flukes, and during courtship the males jostle and bump each other.

Whales are almost completely hairless, save for a few bristles on their heads. Certainly the elimination of hair has improved streamlining, and has reduced the frictional drag. Furthermore, continuously wet hair could not have been of much value in keeping the whale warm. It is also possible that a hairy whale would have been very much bothered by skin parasites which would have flourished in the quiet water between the hairs. However, if this prompted the loss of hair, it was in vain for now the streamlined bodies of humpbacks and right whales are marred by large encrusting barnacles. It is surprising that the barnacles do not completely cover the whales. Perhaps they are scraped off on the bottom, or they cannot flourish during the long migration or in plankton-impoverished waters of the winter quarters. At any rate the parasites are kept partially under control so that much of the streamlined surface is unblemished. 14

If the physical properties of water forced upon whales a common shape, they did at least, by the buoyant effect, free the animals of the need for structural and muscular developments to support themselves against the pull of gravity. Free of this structural problem, whales were able to evolve into the largest mammals which the world has ever known. As they became larger, they had to shift in their feeding to slower and less maneuverable prey. It would appear that the porpoises, which feed on the rapid-swimming, elusive fishes, are small in order to catch their prey. The whales, which have specialized to feed on the jet propelled squids, were able to evolve into much larger whales because they could capture the squid either by stealthy approach or by sucking the squid into the mouth, thus counteracting its jet.



Tailflukes extended for deep dive. *Courtesy Scripps Institute of Oceanography.*

Whale Types

Porpoises and Dolphins

Whales are known technically as cetaceans (pronounced seh-TAY-shuns); so also are the various porpoises and dolphins which are mostly eaters of fish. These are certainly the most numerous of all the cetaceans, making up in numbers for their small size (6 to 8 feet). A few species range between 20 and 30 feet. Porpoises and dolphins congregate around schools of fish. Therefore fishermen are constantly on the lookout for a sight of them. Since not infrequently the porpoises break the surface of the water, leaping completely clear as if for a look around, they are not difficult to locate. Porpoises and dolphins can be seen most frequently in coastal waters where fish are most abundant. The porpoise and dolphin families contain a great many species and it is beyond the scope of this treatise to differentiate or name them all. However, these families include such unique forms as the killer whale, narwhal, white whale (or beluga as it is known to the Eskimos), and the pilot or black whale. Generally one associates cetaceans with the ocean, so it may come as a surprise to find that four dolphin species live in such major rivers as the Amazon, La Plata, Ganges, and the Yangtze.

In dolphins, the mouth protrudes beyond the head as a beak or snout, and in porpoises, the front of the head is blunt or gently rounded. It is impossible to avoid confusion if one uses common names to separate the various whales. Even though the word whale properly covers all the kinds, to some it connotes only the larger species. Such a distinction is wholly arbitrary, and cannot properly differentiate the natural groupings of whales to which zoologists have assigned technical names. It would be impossible to summarize the variety of common names which many of the species have acquired through the centuries. The only solution to this is to refer to the whales by the technical names which connote relationship. (For readers who

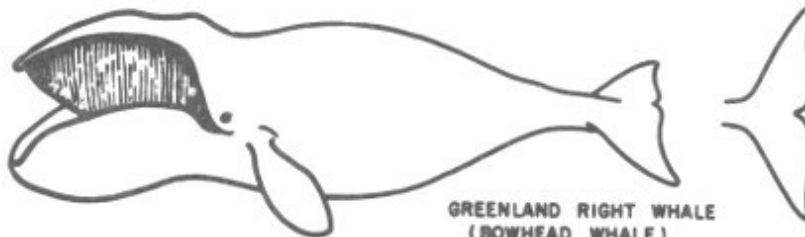
desire this differentiation, a brief listing of the groups and representatives of each are provided in an [appendix](#).)



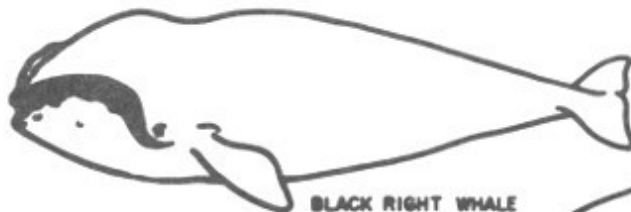
BLUE WHALE
(SULPHUR-BOTTOM WHALE)



FIN-BACK WHALE
(COMMON RORQUAL)



GREENLAND RIGHT WHALE
(BOWHEAD WHALE)



BLACK RIGHT WHALE

BALEEN WHALES

0 5 10 15
SCALE IN FEET



CALIFORNIA GRAY WHALE



SEI WHALE



LITTLE PIKED WHALE
(LESSER RORQUAL)



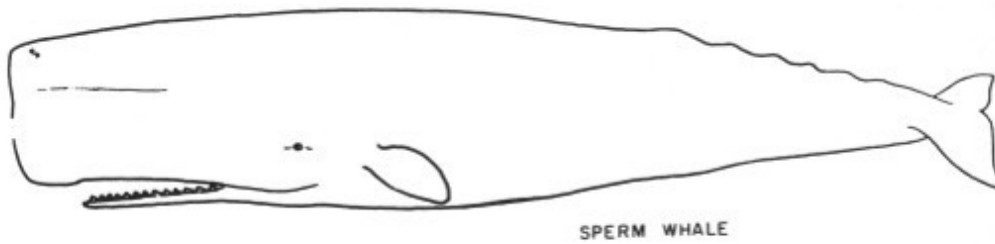
PYGMY RIGHT WHALE



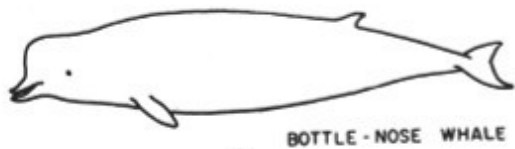
← The Sperm Whale is given here
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BALEEN WHALES

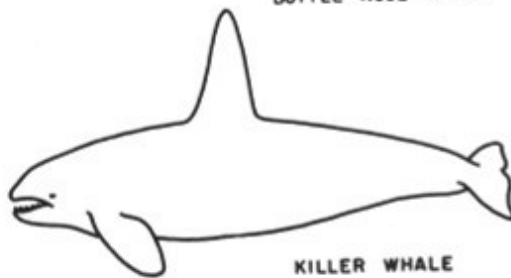
BLUE WHALE (SULPHUR-BOTTOM WHALE)
FIN-BACK WHALE (COMMON RORQUAL)
GREENLAND RIGHT WHALE (BOWHEAD WHALE)
BLACK RIGHT WHALE
SEI WHALE
HUMP-BACKED WHALE (unlabelled in picture)
CALIFORNIA GRAY WHALE
LITTLE PIKED WHALE (LESSER RORQUAL)
PYGMY RIGHT WHALE
The Sperm Whale is given here for size comparison only.



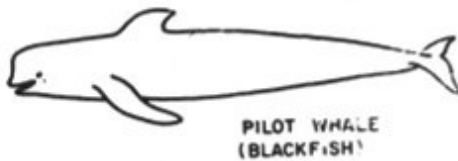
SPERM WHALE



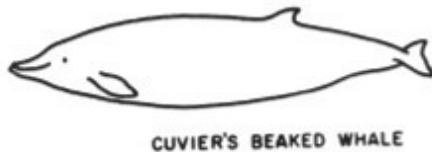
BOTTLE-NOSE WHALE



KILLER WHALE



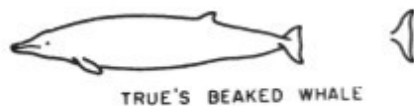
PILOT WHALE
(BLACKFISH)



CUVIER'S BEAKED WHALE



FALSE KILLER WHALE



TRUE'S BEAKED WHALE



TOOTHED WHALES



NARWAL



WHITE WHALE
(BELUGA)



PYGMY SPERM WHALE



BOTTLE-NOSE DOLPHIN



COMMON DOLPHIN



COMMON PORPOISE

TOOTHED WHALES

SPERM WHALE
BOTTLE-NOSE WHALE
KILLER WHALE
PILOT WHALE (BLACKFISH)
CUVIER'S BEAKED WHALE
FALSE KILLER WHALE
TRUE'S BEAKED WHALE
NARWAL
WHITE WHALE (BELUGA)
PYGMY SPERM WHALE
BOTTLE-NOSE DOLPHIN
COMMON DOLPHIN
COMMON PORPOISE

Squid Eaters

The bottlenose whales are nearly toothless, feeding on squid like their close relatives, the sperm whales. Porpoises and dolphins possess many sharp conical teeth on both the upper and lower jaws, although the narwhal which is related to them, breaks the rule by being toothless save for the tusklike canine of the male. In this instance either the right or left tooth elongates to produce an 8-foot spear. The other tooth does not break the gum, and this is the condition found in the female where both are rudimentary and not evident. The bottlenose whales have but a single pair of teeth in the lower jaw, and their relatives, the sperm whales, have 18-28 conical teeth per side on the lower jaw, and these when fully grown may be 8 inches in length. Pockets are provided in the toothless gum of the upper jaw to accommodate the teeth when the mouth is closed.

The decline in the number of teeth in the sperm and bottlenose whales is thought to be the elimination of structures which are no longer useful. Whereas a porpoise's long mouth, bristling with sharp teeth, insures the hooking and retention of a slippery active fish, a small mouth with a few teeth is adequate to crush and slurp down the squid and the weak-swimming fishes of the abyssal depths.

The sperm whale is the largest of the squid feeders, reaching 60 feet. There is a diminutive counterpart, the pygmy sperm whale, which reaches 13 feet. It is exceedingly rare, whereas the sperm whale is abundant in temperate and tropical seas. The beaked whales complete the groups specialized for feeding

on squid. Besides the modification of the mouth, all these whales are noted for their ability to dive to great depths where their food abounds. Not only can they dive to great depths, but they can stay submerged for long periods—up to an hour? Sperm whales have been found entangled in the submarine cables which were known to be on the bottom at a depth of 3,000 feet. It is clear that such feeding habits have opened up vast areas of the oceans to these species.

Filter Whales

The whalebone whales seem to have undertaken two different lines of specialization in feeding: The right whales developed an enormous head with a very large filter plate, whereas the rorquals are much more streamlined with a small filter plate. The ability of the latter to gather food is insured by the pleated throat. The right whales lack a dorsal fin and are 19 decidedly less streamlined. The rorquals have a dorsal fin. There are two species which do not exactly fit in either group. The humpback whale appears to be like the rorquals in that it has a pleated throat and a suggestion of a fin. It is however, a very bulky slow swimming species. The California gray whale, apparently, is intermediate between the two groups and may be thought to be a survivor of the ancestral stock from which both groups differentiated. It has neither a pleated throat nor a fin. The gray whale, like the right whale, has been slow to recover from whaling. It is likely that the populations were never very large. Only the rorquals seem to have the numbers needed for large whaling operations.

Significance of Blubber

Heat Conservation

As whales extended their operations into the icy waters of polar regions or into the cold waters of the ocean depths, they had to develop means of keeping warm. Anyone who has attempted to swim in cold water knows how quickly one loses his body heat and becomes chilled. Whales minimize the heat loss by accumulating a thick layer of fat just below the surface of the skin. The fatty layer, called blubber, not only keeps the whale warm, but it also provides for food storage. It has already been stressed how important it is for whales to survive long periods without eating, so it is likely that the two specializations arose together.

There are, however, extensive areas of the whale which cannot be blanketed with fat and these are the flippers and the large tail flukes. It has been observed that the blood going into these structures gives up its heat not to the outside but to the veins which parallel and surround the arteries. By this anatomical feature most of the heat which would otherwise be lost to the water, is recaptured by the veins which deliver the heat back into the body. Of course, this means that the tissues of the tail flukes and flippers function at temperatures much lower than those found within the body. Here we find that nature was using the principal of the heat exchanger long before man discovered it or put it to work in air conditioning.

Whales are so well insulated that they stay quite warm 24-36 hours after death. Whalers must process the whales quickly, for otherwise, at the elevated body temperature, decomposition proceeds most rapidly and ruins much of the meat. It is possible that the baleen-bearing whales do not cross the warm equatorial waters because they overheat. No one has yet determined whether the newborn young have a sufficient layer of fat to protect them from the cold water, and it has been suggested that whales calve in temperate waters to prevent the babies from being chilled. However,

there are species like the narwhal and the white whale which calve in Arctic waters.

Buoyancy

Another aspect to the extensive deposits of fat is that these tissues are lighter than water and help counteract the heaviness of the whale's body so that with the assistance of the lungs neutral buoyancy is achieved. The fat is accumulated in between the muscle strands, and in fact, in every available nook and cranny.

Food Storage

Much of this fat is drawn upon for food. Whenever a whale is existing on its fatty tissue, acetone is one of the waste products which must be eliminated in the breath. This pungent material makes the breath very strong and noticeable at these times. Certainly among whales, there is no stigma attached to being fat or having halitosis.

Ordinarily fatty tissues only accumulate when there is a surplus of food over the needs of the animal. You might suspect that whales would need to stockpile fat first, in order to remain warm and buoyant, and that growth would be curtailed and accomplished last. However, studies on the growth of whales show that the efficiency of food gathering is so high and food so plentiful, that growth not only continues but at a tremendous pace.

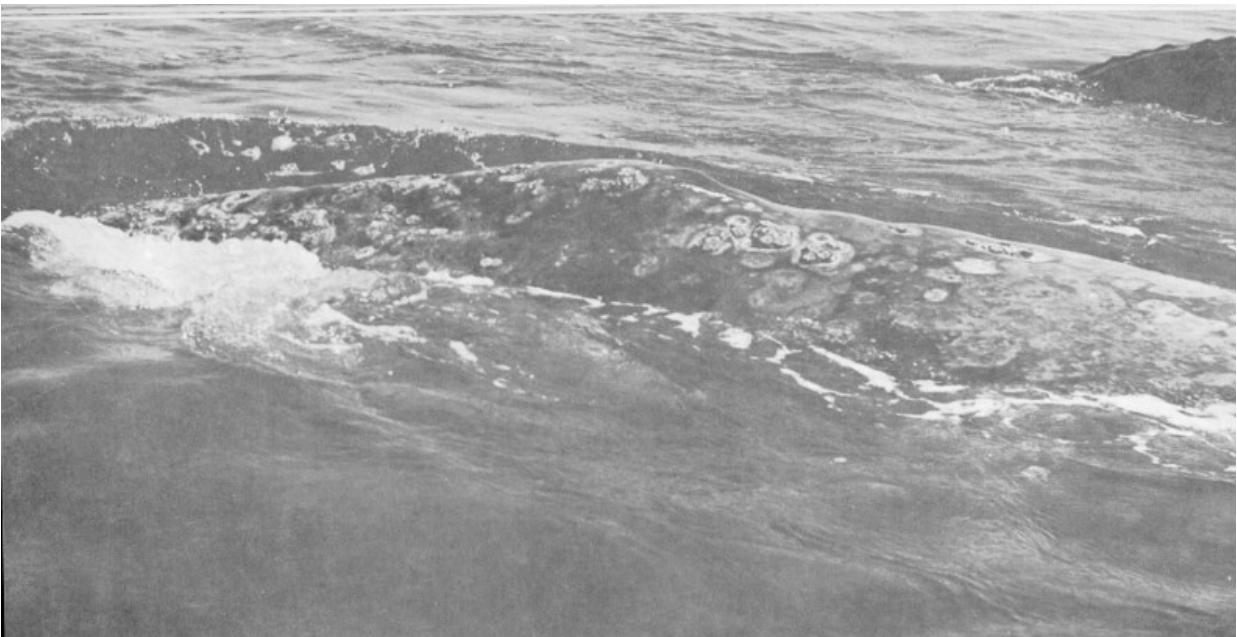
Sexual Maturity

Whales mature sexually between their third and seventh years. Toothed cetaceans attain sexual maturity later than filter-feeding whales. A blue whale is sexually mature at 5 years, whereas porpoises require at least 7 years. Most filtering whales are sexually mature in 2 or 3 years. Whales are not fully grown at sexual maturity, but they continue to grow for years. In most mammals growth stops with sexual maturity. Female whales generally can be expected to produce a baby every other year, for the gestation period is approximately one year. Babies are nursed for about 9 months. At birth the baby is completely formed and active, but lacking baleen, must nurse. A blue whale baby at birth weighs approximately 8 tons, about 1/12th of the weight of the mother. The mother provides the baby with 50 gallons of milk a day. Since the nursing is done under water, and the baby must surface frequently to breathe, the act of nursing is very brief. Muscles in the breasts of the mother force the milk into the baby's mouth in large amounts. The baby will double its length in 7 months, which averages to a daily weight gain of 220 pounds. During all this time the mother must fatten for the winter ahead, and perhaps continue to grow herself.

Life Span

It is not known for certain how long a whale may live after completing its growth. At the present time, commercially important species seldom attain physical maturity before being captured. Many whales which are captured are measured for scientific study. Such measurements also keep the whalers from taking undersized juveniles. Whales apparently do not live to be very old. Fifty years appears to be the best current estimate of a life span.

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Back of a gray whale showing barnacles and barnacle scars. *Courtesy of Scripps Institute of Oceanography.*

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Whale Intelligence

Whales are apparently very intelligent animals. Whalers have remarked how difficult it is to approach whales which have previously escaped. The gray whales were observed actually avoiding the coast after shore whaling had been carried out for a few seasons. Animal trainers have found the toothed whales particularly apt pupils, and these animals are the stellar attraction of the various oceanaria.

Whale Senses

Sight

Little is known, however, about the capabilities of the various sensory organs. Certainly the eyes are very important and are effective under water. It is not likely that the eye is very effective out of water, even though whales do elevate the head out of the water for a look around. The behavior has been appropriately called “spyhopping,” and it is manifested usually in the ice floes. Killer whales are believed to search the sea’s surface and the edge of the icebergs for seals and birds.

Hearing

Whales appear to have very acute hearing. The report of a whaling gun will alarm whales which have previously tolerated the whaling vessel close aboard. Whalers have noted that in very foggy weather whales are much more difficult to approach because of the increased sensitivity to noise. The toothed cetaceans which are gregarious are capable of a great variety of vocalizations. Much of this is ultrasonic to man and it has been suggested that these emissions are used like man’s sonar for finding obstacles and food. Considering the limited range of vision possible in water which is usually hazy or turbid, such a feature would be most useful. There is a continual chatter among members of a porpoise school, or gam, as the whalers call them. The accumulated noise serves as a beacon to which straying members can home when they have gotten out of visual range, which incidentally is under 300 feet. No one yet knows exactly how these animals can produce these sounds without being able to move air across the vocal chords. They do not exhale under water, and yet they are continually noisy.

Smell

The sense of smell is not important to whales, and the organ was abandoned when the nostrils were shifted to the back of the head and modified for diving. Although man may never be able to test whales experimentally for their sensual acuities, it is quite apparent that they are fully aware of their environment. They clearly recognize the environmental signposts which guide them to and from their various areas. Oceanographers are not nearly as adept in knowing where they are on the ocean. Whales clearly recognize their own particular kind, and they do not intermingle. Incidentally, man finds it difficult to differentiate some of the whale species. Because of the rarity of specimens and information, the identity of some species may still be in doubt.

Habits

Filtering whales, unlike the gregarious porpoises and dolphins, seldom school. Even when they appear to be abundant in a limited area, they have congregated for feeding and not for social interaction. Toothed cetaceans, on the other hand, are generally sociable. The sperm whale travels in large groups of females dominated by a single bull whale. The other males have been driven away and the victorious male exercises control of the harem only as long as he wins these contests. Once the dominant male is defeated, he becomes a solitary individual.

Enemies

Killer Whales

Whales, by virtue of their size and speed, are not preyed upon by other animals. The killer whale, however, has the size and inclination. Generally, it is content to capture seals, sea birds, and fishes, and it does not range too far from this prey. There have been isolated reports of killer whales attacking the gray whale. The gray whale is described as being very disturbed whenever a killer whale appears. On the coast of Siberia the gray whale will hide in very shallow water and if cornered is said to go into shock, floating at the surface, stomach up, while the killer whale bites at the tongue and flippers. Perhaps reports like this have been improperly interpreted. It is easy to understand how killer whales would congregate around the catch of the whaler. Recently studies on the loss of hooked tunas from the long-line fishery of the Japanese, show that killer whales are adept at stealing fish along the setline. Additional studies on the natural history of the killer whale must be made before its relationship to other whales can be properly assessed.

Parasites

If whales generally lead a charmed life with respect to predators, they still have their share of parasites both external and internal. Their huge bodies are ideal platforms for the growth of barnacles which have specialized for this unusual habitat. Another very annoying skin parasite is the whale louse, which is a flattish small crustacean which clings by claws to the delicate skin. The digestive tract of the whale provides a wonderful habitat for round worms and tapeworms. Like their host, these parasites are the largest of their kind. Other organs such as the kidneys, liver and lungs are infested too.

Sperm whales suffer from another affliction which is an obstruction of the intestine by a fatty concretion which forms from the bile. Ordinarily these are passed from the digestive tract when small, but if they are retained and continue to grow, an obstruction is possible. The material is a grayish wax which is known as ambergris. It is used by the perfume industry to make permanent blends of various fragrances. Most of this material is obtained by whalers when they process the sperm whale. Only rarely does the material float ashore after the death of the unfortunate producer. Nowadays the value of ambergris is but moderate.

Whale Abnormalities

Whales are found with healed broken bones which must have been incurred by fighting and other collisions. The skin of whales is mottled with scars which were produced by parasites and by fighting. Toothed whales are especially scarred from the raking by the teeth of an adversary during battles for dominance. For at least 5 centuries, man has preyed on the whale. In recent times navies of several nations have been alarmed by unidentified underwater objects which cannot easily be distinguished from submarines. Some of these contacts are produced by whales. The counter measures not only cost the whale its life, but also causes the navies unnecessary expenditures of depth charges and time, and produce considerable tension and anxiety.

Inadequate Knowledge of Whales

Whales have been extensively described both in popular and scientific writings. However, their story is by no means complete or correct. It will be many years before all the information can be obtained on these animals which range the wide oceans where man must study them under great disadvantage. Until man has the underwater mobility and maneuverability of whales, he will have to be content with surmise and interpretations based on limited observations.

There is a great deal known about the anatomy and fine structure of whales. You can certainly admire the work of the early anatomists who persevered in dissecting the partially decomposed carcasses of stranded specimens. What a contrast to the opportunity afforded now to the anatomist who need merely be present on a whaling ship station to receive any part which he wishes to examine. It takes the whaling station 4 to 6 hours to butcher a whale completely.



Close-up of a “spyhopping” gray whale. *Courtesy Scripps Institute of Oceanography.*

Naturally, the whales which are currently the object of research are those which are commercially important and available. The California gray whale which would be convenient for study is no longer taken commercially. Permission would have to be obtained from the International Whaling Commission to secure even a single specimen, and a scientist could not easily obtain a specimen without the assistance of whalers. Fortunately, we know a great deal about the species as a result of an extensive monograph prepared nearly 100 years ago by Captain C. M. Scammon, who was among the first to whale this animal commercially. Professor Edward D. Cope described the species, giving it the name *Rhachianectes glauca*, after he examined a skeleton which had been shipped to him at the Philadelphia Museum.

Roy Chapman Andrews of the American Museum prepared the first detailed description of a living specimen which he found at a Korean whaling station. At the time of his report the species had not been seen for many years

and had been believed extinct. More recently Russian scientists published considerable material which they had obtained from Russian whalers who had continued to take an occasional specimen during the thirties of this century.

They made a detailed study of the food of this whale, and they clearly showed that the herd which summered in the Sea of Okhotsk and wintered off Korea did not intermingle with the Siberian herd which wintered in the lagoons off Lower California. For the last 20 years or so, the species has enjoyed protection from whaling on an international basis. Since 1947 whaling scientists of the Canadian and American fishery agencies have published data principally on the recovery of the California herd, and to a degree have filled in some of the gaps in its life history. There is still a great deal to be learned, and it may be that some of the published observations will have to be revised or discarded. It is likely that the California gray whale is quite similar in many of its biological details to those which appear to be general in the commercially important species.

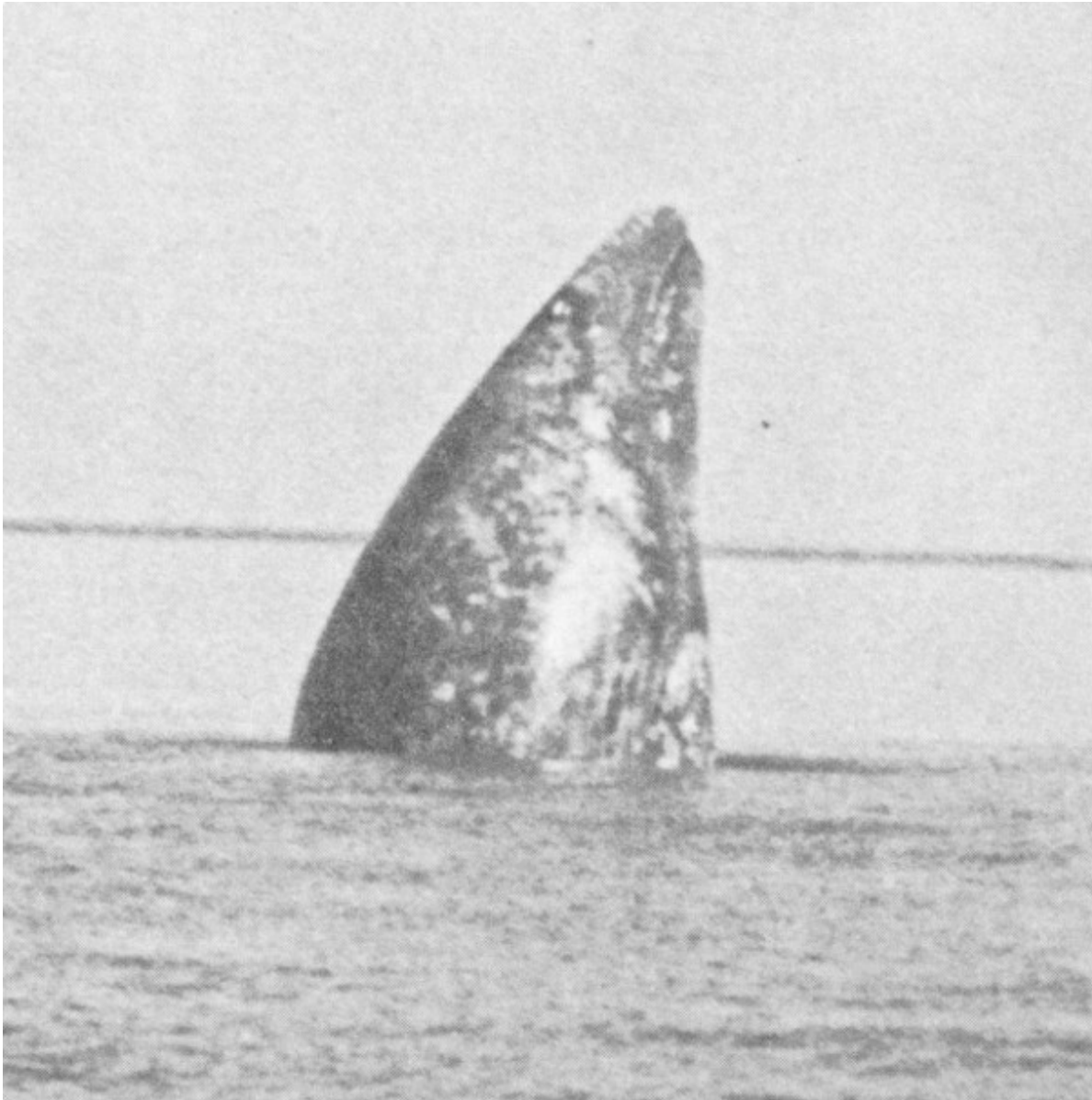
California Gray Whale

Evolutionary Place

It should be kept in mind, however, that this whale is the most primitive of the living baleen-bearing whales, and it seems to represent the point in the evolutionary sequence from which the two main groups diverged. The California gray whale is the sole survivor of a family of whales which in past eons was represented by many species. It has a smallish head with a limited filtering mechanism. Its dependence on foods which are found only near to shore is considered further evidence that it is primitive. Likewise its need for harbors to calve suggests that it has not lost its dependence on the nearshore habitat, even though it can safely cross the wide Pacific. The extremely restricted and separated distribution pattern of the whale is typical of an old, primitive species which persists either in its preferred ancestral home, or which occupies less desirable habitats into which it was forced by the more successful species. Recently gray whale remains were found in the Netherlands, and it is possible that it was exterminated in European waters by prehistoric whaling prior to the Ice Age.



Gray whale breaching. *Photograph by T. J. Walker.*



Another gray whale “spyhopping.” Notice the narrow tapering head, typical of the California gray whale. Courtesy of the Scripps Institute of Oceanography.

Geographic Distribution

Today the species is restricted to two distinct populations existing on opposite sides of the Pacific. One herd summers in the Sea of Okhotsk, migrating southward to Korea where, in the open bays, calving takes place in January, February and March. The larger population summers off the coast of Siberia and Kamchatka, migrating clear across the Pacific Ocean to

California and then south to the breeding lagoons near Bahia de Sebastian Viscaïno, situated on the outer coast of Baja California. A few whales 30 have been noted around the tip of Baja California and in the gulf itself.

These are part of a small group which calve in lagoons along the mainland of Mexico just inside the Gulf of California. It has been suggested that the recovery of the species was possible because these lagoons were unknown to whalers. It is, however, unlikely that the whalers continued to whale until they had completely wiped out the populations of the main lagoons. The inability of the species to support intensive whaling suggests both a limited population and an animal with a low reproductive potential. Since 1850 there have been three brief periods of whaling, each separated by about 20 years. Each succeeding period of whaling has been shorter, suggesting that 20 years is not a long enough recovery period.

Shore Habits

Generally speaking, shallow water is dangerous for the larger whales, other than the gray and right whales. Strandings of the other species do occur, which can be interpreted to mean that these whales are not afraid of shallow water. However, when they strand they die. The cause of stranding is difficult to determine. Perhaps the victim is sick and drifts ashore. Most strandings occur during severe storms which quickly obscure the bottom features by stirring up the sand and mud. The stranding of the pilot whale is most dramatic because entire groups will come ashore as if afraid to go back to sea. If man intercedes and helps free them, they still refuse to escape. The cause of the panic and subsequent action is unknown, although it is unlikely that the action is mass suicide as so colorfully described in newspaper accounts. Most other strandings suggest that the whale was out of its normal habitat.

The gray whale, however, has no difficulty in this dangerous region, and in fact they negotiate the treacherous turbid channels of the lagoons, notwithstanding the strong tidal currents which flow through them. They have also been observed rolling in the surf in water barely deep enough to float them. Should stranding occur, they seem to realize that it is only momentary as another swell will come along to free them. For the other

species, stranding produces fright and struggling which only exhaust them and lodge them deeper into the bottom. Death usually results from overheating, although it is noted that a stranded whale finds it difficult to breathe normally because the weight of the body out of water crushes the chest cavity.

At the present time it is difficult to understand why the lagoons are used by the gray whale for calving. It may be an ancestral behavior trait which has been inherited. It is most likely, however, that the quiet waters facilitate birth and nursing during the time the baby is gaining strength for the long migration. The observation that the whales may roll and rub themselves against the muddy banks has led to some conjecture that the lagoons help control the skin parasites. 31

You may wonder whether the lagoons, which are limited in number and in extent, may control the size of the population. Lagoons are subject to change in size and depth through geologic processes, and they may be created or destroyed. The lagoons which are used by the whales are remote and seldom visited by man. You wonder whether the whales would continue to use them if they were also used much by man. The California gray whale has not used San Diego Harbor for over 100 years. Perhaps San Diego Bay was used because the other lagoons were then overpopulated. Certainly the total population of whales has never again reached the numbers which existed before the whalers started.

Scientific Description

It has already been mentioned that the gray whale was described by Professor Cope in 1868-69. The scientific name by which he tagged it, *Rhachianectes glauca*, describes the fact that the whale is blue-gray in color, and that it frequents rocky reefs. Because the species did not resemble any other whale, he placed it in a separate family, the Rhachianectidae. Cope was not aware that paleontologists had found and described fossils of whales which were very similar to the gray whale. Of course, the paleontologists working in Europe did not know that there was a living representative of these fossil whales which they had named *Eschrichtius*. However, this

oversight was finally realized by Dutch cetologists Van Deinse and Junge, who in 1937 corrected the mistake by placing the gray whale in the genus *Eschrichtius*, discarding the old genus name *Rhachianectes*. If you wish to search the scientific literature on the gray whale prior to 1937, you must look for it under the old name *Rhachianectes*. This is not the first or last time a scientific alias will be produced; yet this one serves to dramatize the fact that the gray whale is truly a living fossil, which is a reminder of some of the evolutionary steps over which the other whales passed to gain the structural and behavioral features needed to occupy the many areas of the oceans.

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Infra-red photograph of a gray whale. Note the lingering spout and the blow holes at the forward end of the surfaced whale. Photograph by T. J. Walker.

Reaction to Whaling

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The California gray whale is known by other common names, such as “Koku Kujira” of the Japanese, which means “the devil fish.” This indicates the cleverness of the gray whale and the dangers associated with hunting it. Whalers universally considered this species the most difficult to whale and the only one which might charge the boat. Gray whales certainly seemed to

learn quickly the dangers to them from whaling. Captain Scammon considered this species the most interesting, giving it top billing in his book on the *Mammals of the Pacific Coast*. He reports that it was standard practice in the lagoons to kill the baby in order to lure the mother into harpoon range. In these encounters, the whalers showed great courage standing the charge of the angry mother, as they took a chance of losing their boats and their lives. The females of other whale species have a strong maternal instinct, but none of them seems to be so consistently willing to fight for its young as the gray whale.

We still do not know much about the behavior of the gray whale or the other filtering whales. At present, the people who are best informed on this subject are the whalers. They can ill afford to waste valuable time trailing whales which are too wary to be approached. In general, feeding whales are easiest to approach, being intent on feeding and completely oblivious to the approach of a whaling vessel. At other times, the whales are disturbed by the least noise. Ordinarily, once a whale is harpooned, it attempts to escape its tormentors, rather than turn and demolish the boat, which of course, would have put an early end to whaling. In the instances where a whale has rammed a boat, it is usually an accident, occurring during the death flurry. However, there are exceptions, and of course, these make whaling hazardous.

How Do They Sleep?

One question about whale behavior that remains somewhat unanswered is—how much rest do whales require? There have been indications that the California gray whale in the lagoons of Lower California may rest a good deal. However, the question of sleep during the long migration or on the feeding grounds, has not been answered, and as yet attempts to determine this have been hardly more than preliminary tries. In order to succeed 34 at this venture, observers must be able to remain close enough to the whales at night to be within sound of the spouts. Limited visibility at night, and the amount and direction of the boat's drift, make it difficult to keep contact with a whale spouting only once every 4 to 15 minutes. The scientists who sail with the whalers to study whales are too busy with the

daily catch to consider this problem of how much a whale sleeps. Occasionally, a whale is struck by a ship during the night, and this is considered as indirect evidence that the whale was asleep on the surface.

Migration Groupings

The California gray whale is not believed to be particularly gregarious. The groupings observed at the lagoons are thought to be due to the limited areas which are available for mating and calving. There is some indication that they form groups during their migrations. Russian observers say that early in autumn groups gather and start out on the migration. The Japanese have noted groups of moderate size remaining together during the crossing of the Bering Sea. As yet no one has attempted to trail a group of grays along the entire migration route. Not only would this establish the route, but might also add to our knowledge about their behavior en route.

At San Diego three categories of migrants have been observed on the southward migration. During the early and middle periods of the migration, many individual whales are seen apart from the group. These are believed to be pregnant females. They travel at about 4 knots and surface frequently, about every 3 minutes. The migrating groups of three to five are believed to be nonpregnant females accompanied by one or more males. These are already courting, and it appears that the males are jousting for the attention of the females. At any rate, the migration is less hurried, and the individuals spend a great deal of time on the surface, blowing and milling about. Such groups are seen principally during the middle part of the migration, and they are easiest to sight. They afford the spectators at Cabrillo National Monument quite a show. Not infrequently, these groups put on a display for an hour or two. It is difficult to observe all the details of courtship, but it is believed that during mating the animals lie quietly in the water. On the migration, the courting whales roll over in the water extending the flippers alternately like spars. Mating generally takes place near the lagoons and also in the lower ends of the lagoons. A few matings have been observed at Point Loma.

Breaching

Occasionally the gray whale will burst out of the water at about a 45 degree angle, and will turn over, falling back with a tremendous splash. This is called breaching, and its significance is not known. Although it is considered part of the courtship behavior, this is difficult to reconcile with the observation that it is more frequently displayed by solitary migrants. Whales may breach more than once, and in fact, one was observed to jump repeatedly, as many as 25 times within an hour's passing of the Point Loma area. Here one or two migrants will breach each week. Unfortunately, the act takes place so quickly that you only notice the final splash, at best obtaining only a quick glimpse of the whale's massive body. However, this sight repays many hours of patient watching.

Toward the end of the season you see that the migrants are smaller and closer to shore, frequently just beyond the surf. These are believed to be last year's young, making the trip alone for the first time. They not only swim more slowly, but they also seem to wander into the kelp and into every irregularity of the coast. Those which enter the kelp may raise the head vertically out of the water in an action called spyhopping. They are obviously trying to find a passage through the kelp. The fact that the gray whale does not spyhop regularly on its migration along the shore must mean that its orientation is based either on the sound of the surf or the depth of the water. Spyhopping is apparently employed by the whales in the lagoons, and it has been observed frequently for the other species in the ice floes of the polar seas.

Whaling

The role which whaling has played in man's conquest of the oceans has never been given full credit by historians. Initially, man was content to fish along the shore, and then gradually he pushed out on the waters in sight of land. Here he learned to be a skilled fisherman, and he began to hunt the coastal whales. Man gradually developed the boats, gear, and the know-how, called seamanship, and gained the courage to venture farther and farther from land. Even as late as the 15th century sailors were extremely superstitious and believed in sea monsters. There was, of course, a certain basis for these imaginings. Periodically, a large whale would strand or float ashore and it is easy to understand how these tremendous and misshapen remains could have kindled the imagination. Whaling was an extremely hazardous profession, calling for the utmost in nautical skill. Throughout the development of the maritime resources of some European states, whaling was one of their most profitable businesses. Even though a great deal of shipbuilding was initiated by man's speculative desire to find new lands and to conduct trade, an equal amount was for whaling vessels. From the Middle Ages to nearly the end of the 19th century, oil from whales illuminated some homes and streets. Whale meat was an important source of protein for peoples who had not yet developed extensive animal husbandry.

For over five centuries whales have been under attack by professional hunters, and long before that by the aboriginal peoples of many lands. The reason whales have withstood this hunting pressure better than land animals is due to the fact that their ocean habitat has not been invaded by man for agriculture, and in the days of sailing ships and hand harpooning, the whale was a formidable prey, far more so than in our present day of motorization and automation. The tremendous area of their habitat made it difficult and time consuming for man to find them, and his ignorance of their migration routes and the oceanographic features which cause whales to disperse and congregate gave the whales some protection. Technological advances, powerful harpoon guns and totally mechanized factory ships in the whaling

industry have swung the advantage to man, and whales are now in peril of extinction as is the whaling industry itself.

Whalers as Explorers

History tends to stress the great explorers who discovered new worlds and the larger islands of the Pacific. However, ships cannot with full safety, ply the seven seas until every island and shoal has been found and charted, and ships cannot safely venture near any shore until the adjoining waters have been systematically charted. The principal discoverers of these danger spots have been the whalers who patiently combed so much of the oceans in their search for whales. Every island has on it the calling cards of Whalers who put ashore for water, food, fuel, or just to stretch their legs and to satisfy their curiosity. This isn't to say that they had the time to make hydrographic surveys, but at least their log books defined the areas where surveys would eventually have to be made. Whalers lived adventure, and danger was their common lot. It is not surprising that much of this adventure would find its way into the hearts of men through yarns and tales such as *Moby Dick*.

Prehistoric Whaling

Whaling was practiced long before historic times. The coastal Eskimos of the arctic Americas and of Greenland and Spitzbergen practiced whaling. Whales were approached in skin-covered boats and even from the edge of ice floes. These peoples are differentiated ethnically on the basis of the variations in their whaling techniques, in their whaling gear and in their religious preparations for whaling. It took great courage to whale in such frigid waters where an overturning meant death. It is not surprising that these peoples looked to supernatural deities who required a very complex ritual to insure a safe and successful hunt.

The principal method of hunting involved coming up alongside the whale so that the man in the bow, the whaling captain, could spear the chest. If he were lucky enough to drive the shaft between the ribs, the lung could be

pierced and the whale would be mortally wounded. Each time the whale surfaced the boat would attempt another approach and lancing. The stone tip of the lance was detachable, and when freed, led to an inverted sealskin float by a long line of twisted sinews. These floats marked the location of the whale during submergence. Eventually the whale, weakened by loss of blood and by fright, would die and then came the chore of hand pulling the whale ashore for feasting and reprovisioning the village's storehouses. The captain of the boat and his crew were considered the most skilled and important of the hunters. Whaling was an ancestral skill which was passed on generation to generation with great care. The villages could ill afford to lose their best men, or the equipment which was difficult to make.

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There are interesting variations to this method. Aleutian islanders prepared a poisonous coating for the harpoon tip which was sufficiently toxic to kill the whale. This was prepared from the deadly aconite by a secret process, depending upon the concentration of the alkaloid poison from the root. Eskimos from Greenland and Spitzenberg dried another secret coating on the harpoon tip which consisted of a bacterium which produced fatal blood poisoning in the whale. In either method the whalers waited for the bloated carcass to surface and then they hauled it ashore for eating. Apparently the spoiled meat had no ill effects on these peoples, but of course, much of their food was spoiled during the summertime.

Much of the religion of these primitive peoples centered around the need of giving the whaler and his crewmen the necessary courage and confidence to perform this job. Religious ceremonies were held weeks before, and again after, every hunt. Some of the deities were whales, and whales also figure predominantly in many of their folk tales.

The European and American whalers came in direct contact with the Eskimos competing not only for whales, but also for seals, walrus, and fur-bearing animals. Today, Eskimos use harpoon guns and sturdy wooden whale boats. However, the entire village still turns out to drag a whale ashore. Each village kills only what it requires for food and oil through the long arctic winter. They hunt the arctic bowhead whale, although the smaller white whale (beluga) is also hunted. The bowhead has never recovered from commercial whaling, even though it has been protected for years by the International Whaling Commission. The regulations however, do not apply

to the aborigines, but their needs should not have prevented the whales from becoming plentiful again.

The Indians who occupy the coastal waters of south Alaska and British Columbia are excellent salmon fishermen, but do not whale, even though they possess magnificent boats suitable for this. However, at the entrance to Puget Sound and south along the Olympic Peninsula live Indians who hunted whales as recently as 50 years ago. They whaled in much the same way as did the early Eskimos, with detachable stonetipped lances and sealskin floats. It is believed that the whale they hunted was the California gray. Boats were hewn from the massive trunks of cedars. From the bark and wiry branches, a tough flexible line was woven, strong enough to withstand the terrific pull of the whale. These Indians repeatedly lanced the whale until its death. The generation of Indians who knew this skill is gone, and the present generation only remembers vaguely the tales of their grandparents. The Washington State Historical Museum in Seattle has managed to interview the old-timers and collect their gear which is now on display there.

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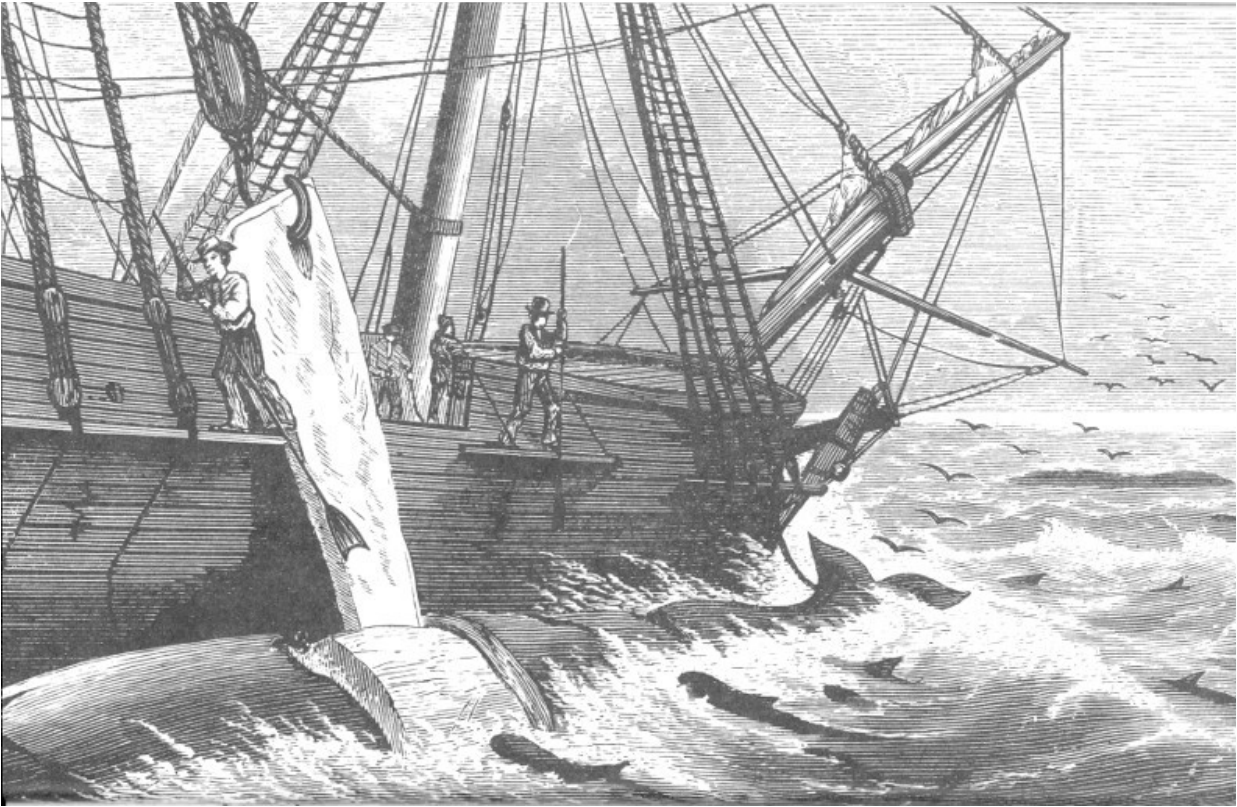
The Kamchadal and Chuchee Eskimos whose villages looked out on the summer grounds of the California gray were not inclined to whale. They did, however, feed on any which happened to wash ashore.

Historic Whaling

Commercial whaling began in Spain during the Middle Ages. At that time the black right whale was plentiful along the coast of the Basque province, and it provided an ideal quarry because it could be sighted from lookout towers along the coast and it was not difficult to approach. Equally important, these whales floated at death so they could be pulled ashore. They were prized for their oil, meat, and the baleen which was sold throughout Europe. The Biscayan whalers gradually reduced the local population of whales and began to range farther in their search. They came at last upon the haunts of another right whale which frequented the ice-filled seas of the North Atlantic. Here there was not a satisfactory method to preserve the meat so this was thrown away.

There is an extensive and fascinating history of this early whaling but only the briefest outline can be mentioned here. Throughout the story there is a constant search for new whaling grounds to sustain operations and to meet the increased demand for whaling products. As the industry expanded it was forced to capture less desirable species because the other species could not hold up against the onslaught.

Other nations took up whaling and soon the English and Dutch had large fleets of ships manned by Basque whalers. These ships were based in Greenland and Spitzbergen. The irregular coastlines of these wind-swept lands provided many harbors in which shore whaling stations could be situated. The camps were well constructed, and daily during the arctic summer, the ships put out for whales in waters which were rough, cluttered with icebergs, and shrouded in fog. Naturally, under such difficult conditions, there was great loss of ships and men. The companies were not friendly with one another and periodically raided the shore camps of each other, stealing and burning supplies and buildings. Under normal circumstances such activities would have been provocative of war. However, since communications were poor and the countries concerned were preoccupied with colonizing America, the situation took care of itself, for in time the whales diminished and the whaling companies failed.



Cutting-in, from the book *Nimrod of the Sea or The American Whaleman* by Wm. M. Davis.
Courtesy San Francisco Maritime Museum Association.

Soon after this the colonists in America wrote to relatives in Europe about the richness of the country, not the least of its resources being the whales which spouted just off shore along the entire Eastern seaboard. Whaling companies were formed and these were concentrated along the New England coast. After the separation of the colonies from England, whaling developed locally with the ships which had been engaged in commerce between the colonies and the mother country. It did not take long to exhaust the populations of whales, so the Whalers began to range afar.

This led to the discovery of the haunts of the sperm whale in the open oceans of the tropics and subtropics. Yankee ingenuity developed a new kind of whaling in which the whales were brought alongside the base ship for processing. It took a great deal of skill to strip the blanket of fat from a whale which was three-fourths submerged and pitching and rolling on the restless surface. The blubber once removed, was hauled aboard and cooked in kettles to free the oil which was then placed in barrels and stowed below. This form of whaling was called pelagic whaling. It was not uncommon for

ships to be away from home port for a year or two before the casks were full. This whaling was dominated by the Americans, and it contributed much to the prosperity of New England, providing much of the capital which later financed its manufacturing developments. This was the golden era of whaling so superbly described by Herman Melville in *Moby Dick*. Here is a real yarn about whaling written by a Whaler who later became a superb author.

The discovery of petroleum in Pennsylvania brought to the fore new illuminating fuels, kerosene and natural gas which were much cheaper than whale oil, so the whaling industry declined rapidly. This happened at the same time steam ships were replacing sailing ships, so it is not surprising that the beautiful sailing vessels which had been the mainstay of whaling were brought home to every little coastal village in New England to rot away, sad memorials to better days.

Modern Whaling

By the 20th century whaling was again possible because of new uses for whaling products. Chemistry has succeeded in finding new uses for whale oil. Fat derivatives such as soaps, margarines and a few other products owe their abundance and low cost to the whaling industry. Packing house technology has been applied to the butchering of whales so that the entire animal has many uses. Tasty meats are finding their way into the diets of many countries and the tougher, less tasty parts are ground and used to feed cats and dogs. Farms that raise mink and other mammals for the fur trade are heavy buyers of this source of meat to feed their animals. The remaining parts, blood, bones, and meat scraps, are dried and ground into nutritious meals which are used by the poultry and livestock industries as supplementary feeds. Currently each whale produces over \$5,000 worth of products.

The heart of the processing operation is the pelagic factory ship which is nearly as large as an aircraft carrier. They are supplied by huge tankers which carry off the oil and by refrigerated ships which return the meat to port. In the 20th century the United States has not seriously participated in

whaling and has not a single fleet in competition with the other countries which now have billion dollar investments in ships and gear. For the past 70 years, the leader in the whaling industry has been Norway. The industry is very competitive however, and England, Japan, Netherlands, USSR, and Germany have big investments.

The Norwegian lead was due principally to Svend Foyn who invented the harpoon with an explosive head. The harpoon was fired by a deck-mounted cannon and it had sufficient range and accuracy to permit killing those whales which heretofore were too fast and wary to be handlanced. The harpoon head had a time delay bomb which insured that the explosion occur deep in the vitals, and that the whale was made fast to a manila line so that it would not be lost. The cannon is mounted in the bow of a small fast ship known as a killer boat. These little ships operate for 1 or 2 days away from the factory ship in search of whales. The gunner who is responsible for firing the cannon is the most important member of the crew and on his experience and skill depend the entire operation. If he misses or fails to kill the whale, it will alarm the other whales and a day or two of searching may have been in vain.



Whalebone drying; from the book *San Francisco Whaling* by Pacific Steam Whaling Co., San Francisco. Ships in picture—the *J. D. Peters* on the right and the steam whaler *Orca* on the left.
Courtesy San Francisco Maritime Museum Association.



Trinidad, Calif., whaling shore station; two humpback whales on the platform. *Courtesy San Francisco Maritime Museum Association.*

The whaling season is extremely short and as soon as the yearly quota 45 has been reached, all countries must stop. Naturally every country tries to get as many whales of the total as possible. Once a whale is killed, it is hauled alongside, and the body cavity is inflated with compressed air to make the whale buoyant. It should be remembered that the whales which are now being processed are the “wrong” whales and they would otherwise sink. The whale is then flagged, lighted, and a radio transmitter secured to it, and it is cast adrift while the killer boat continues to search for more victims. At the end of the hunting period, the killer boat retraces its route, picking up the whales it has killed. These are secured to the sides of the ship and brought back to the factory ship for processing.

Naturally, the technological improvements which make whaling profitable again would have been meaningless without the discovery of new whaling grounds. About 1900 the seas around the antarctic continent were found to be teeming with whales. These were principally three species—the humpback, the finback, and the blue. The blue whale is distinguished by the fact that it is the largest mammal on earth, reaching sometimes 100 feet in length and weighing over 100 tons. It is the species preferred by whalers but it is not as plentiful as its close relative, the finback, which is the second

largest species, ranging in length from 60 to 80 feet. The humpback whale has never been too abundant, but along with the rorqual, or Sei whale, it has provided an extra dividend for the business.

The extreme richness of the antarctic whaling grounds is due to the fact that the entire perimeter of the continent is open to oceanic waters, allowing whales free access to a tremendous area of fertile waters. The limited richness of the arctic whaling grounds is due to the fact that the Arctic Icecap is flanked principally by the land masses of Eurasia and North America and the whales have access only to those portions opening from the Bering Sea and to a small area in the North Atlantic. Whales cannot swim too far in along the coast of either mainland because the period of open water is brief. Even in the antarctic whales tend to work near the ice floes and occasionally get caught by the sudden closures of the ice.

It is too difficult to estimate the number of whales which were taken from the arctic during the many seasons that whalers worked there. However, it cannot compare with the numbers which are realized annually from the slaughter around Antarctica. As many as 45,000 whales have been taken in a brief whaling season of less than 2 months. Contracting governments under provision of the International Agreement For Regulation of Whaling enforce the regulations as regards the number taken and there is a penalty if a juvenile whale is killed. Nursing mothers are also protected. The regulations are made by the International Whaling Commission and the number of whales which can be taken each year is determined from the statistics of the populations and these quotas are recommended by the Commission. It is hoped that by these regulations and studies, the industry can continue to prosper. The problems and activities of the scientists, the majority of whom work for the various governments' agencies, will be considered shortly.



Blubber ready to process at the Moss Landing, Calif., whaling shore station, photographed 1919.
Courtesy San Francisco Maritime Museum Association.

Example of Unregulated Whaling

California Gray Whale

No account of whaling would be complete without mention of the shore and lagoon whaling of the California gray whale. There were 11 shore whaling stations scattered along the coast of California from San Simeon to San Diego, which had two stations. This whaling dates back to the gold rush days of California. It was not an uncommon experience for a whaling ship to come from the east and then to lose the entire crew in San Francisco before they could get on into the North Pacific to whale.

Although the principal profit for these whalers came from the arctic bowhead, it was customary to return to San Francisco to resupply in the autumn, and then to whale the California gray during the winter migration. Of course, this was whaling made to order because there was no need to search. The shore stations were able to process more whales in a day than a vessel could in a month of routine searching. Captain Scammon followed the migrating whales to the lagoons where he found them calving. These lagoons became the headquarters for the whaling ships which found whaling even easier than they had along the migration route. One of these lagoons still bears Scammon's name.

It was possible to watch the killing of whales from Point Loma, and from this vantage point the whaling boats were directed by flag signals to their quarry. The whales, once killed, were towed, tide permitting, by the hard rowing of the whaling crews back to Ballast Point. The numerous California shore stations have long since closed and rotted away and the grounds occupied with new enterprises. Most of the details of this era are lost, forever or buried in yellowing manuscripts and newspapers.

Although the gray whale may have been easy to find and to approach, whalers noted that those which escaped had learned the implications of

whaling and thereafter were difficult to take. Lagoon whaling was particularly hazardous because the mothers frequently would charge the boats. Much of the time of the ship's carpenter was used repairing the damaged boats. Of course, the exposure of the breeding and calving grounds to whaling had a catastrophic effect, for the whales soon disappeared. It is decidedly difficult to determine from the limited records which remain just how abundant the gray whale was before whaling started. One is likewise puzzled by the report that a thousand a day passed San Diego. If this estimate had been made by an experienced whaler, it might be reliable, but if it were made by a layman, it could be most inaccurate. It is doubtful that anyone would have actually counted the whales for even an hour, so that the figure is probably an exaggeration.

The decline in the abundance of the gray whale occurred during the period when whaling was nearly over because of the low price of oil. Two brief subsequent attempts to whale this species produced only a small return of individuals, leaving the species virtually extinct.

Gray Whale Protected

In 1937 the International Whaling Commission declared the gray whale a protected species and forbade its members from whaling it. The population has recovered noticeably in the past 10 years, but it is too early to predict when it will level off. In 1959 it was estimated that the total population was somewhere between 3 and 5 thousand. It is hoped that the whaling nations will be content to honor their gentlemen's agreement to protect it until recovery is complete.

Value of Gray Whale

Esthetic

Already the whale has been viewed by hundreds of thousands of Californians and tourists who have been thrilled to see such an animal. Currently each season excursion boats are transporting visitors on a 2-49 hour excursion practically out onto the backs of the whales. It is hoped that the pleasure which it offers man will transcend the brief monetary profit which might accrue to a few whalers. It would indeed be shortsighted to let commercial interests deprive man of such a unique whale which can be enjoyed so easily by so many.

Scientific

There is real need to protect the gray whale for scientific studies. The principal problem in the study of populations of whales is to learn how large the population is and what are the various forces which limit its size. Such knowledge would make it possible for the industry to decide how many whales could be taken each season without reducing the population. The concept of taking only those whales which would overpopulate the habitat is an attractive one, and this would be in the long run, most profitable and stabilize both the whale population and the industry. Such knowledge also would prevent the overexpansion of the industry, which would also reduce the profit and hasten the decline of the resource.

Up to now, the studies which have been made to gain this knowledge have to be made on populations which are under severe exploitation. The researcher can never be certain that his findings are normal and useful in estimating the total population. The California gray whale provides him with a control population which is not under predation by man. Furthermore, this

population becomes concentrated during breeding and calving and briefly during migration so that counting the total population is feasible.

Commercial

There is no doubt that the whaling industry will resume the harvest of the California gray whale once it has recovered. This is indeed unfortunate as the species is neither abundant nor large enough to provide much profit. Certainly history will repeat itself and the grand spectacle of the California gray whale migration will require another thirty or more years to repeat itself. The future status of the California gray whale may not necessarily have the same fortunate outcome, as the whaling industry at the present time is much more thorough. Furthermore, competition between the various whaling nations makes adjustment of regulations slow and cumbersome—much too slow for a species so limited in numbers and so accessible to complete extermination.

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The bone pile at the Moss Landing, Calif., whaling shore station, photographed 1919. *Courtesy San Francisco Maritime Museum Association.*

Already the Russians have expressed proprietary interest because they control the summer feeding grounds. Likewise, with the same reasoning, the Mexicans control the breeding waters. It would appear that the American interests, limited to two small shore stations at San Francisco, must be content to secure those whales which will escape the efficient Russian and Japanese fleets which could, at the most opportune moment, secure the species before the beginning of the long migration. At any rate, the species can only serve as an extra species in areas where whaling is already marginal because of limited stocks. It is hoped that eventually the whaling interest will remove the gray whale from whaling, leaving it as a living memorial to whaling. 51

Natural Population Controls

Conservationists should remember that nature has wisely designed each species of plant and animal with a built-in margin of safety; namely, a surplus of young which will repopulate the species from periods of extreme adversity. Yet these surpluses must somehow usually be eliminated lest the species overpopulate its habitat and destroy its own sustenance. The controls which limit the population are many and the population existing from year to year is the statistical average of these many controls. Diseases, predators, and other adversities are necessary evils, which are in the final analysis blessings in disguise. Whales, too, have their checks, although we are a long way from knowing their relative importance.

Perhaps the most critical moment in the life of the whale is birth, because the newborn whale must surface immediately or suffocate. Any abnormality in the birth process or weakness on the part of the infant may cause its loss. Inasmuch as a whale calves only every other year, the loss of a baby is serious, and especially so because a whale produces but a single calf. Less than one per cent of whale births are twins which is about the same frequency as for humans. A careful examination of the shores around the breeding lagoons reveals that a few babies are lost at birth.

Current Scientific Inquiry

Life Expectancy

Life expectancy is another very useful statistic for population studies. The whaling industry has no easy way of finding this out because whaling never gives an individual whale the chance to attain old age. Perhaps we can eventually determine life expectancy for the gray whale if it can continue to be protected from whaling. By counting the number of babies produced each year, and knowing the life expectancy, We might indirectly obtain the number of whales which die before they have realized their life expectancy. This number should represent the surplus on which whaling could be based. A further study would be necessary to determine what percentage of the surplus could be diverted to the whaling industry.

Age Determination

The problem of aging whales is an important one, and one to which much thought has been given. Actually, it is desirable to know at what age sexual maturity occurs; at what age physical maturity; at what age reproduction is no longer possible, and lastly, the age at death. In addition, we need to know the gestation period and the average number of pregnancies of which the female is capable. Many of these statistics can be obtained from the examination of a freshly killed whale. Such age determinations are made on every specimen which comes aboard the factory ships and also at the shore stations. There is some doubt as to the accuracy of some of these. Physical maturity is delayed in whales for many years, and it is judged complete when the cartilaginous end plates fuse to each vertebra (backbone). After fusion, further lineal growth is impossible and after this time whales have been noted to shrink slightly each year.

The other structures which a biologist collects for aging are the ovaries of the female whale. These are sliced across into half-inch slabs, and the number of exposed grayish masses are counted. There is one of these structures produced for each pregnancy, and they persist as scar tissue throughout the life of the whale. These structures are functional during pregnancy and nursing and are responsible for a successful pregnancy. A count of these gives the investigator the total number of offspring which have been produced. If one assumes that the whale has not missed an opportunity to produce a baby, the count of these gives the age, 53 assuming one knows the age of sexual maturity and the maximum length of bearing.

In recent years other structures have been tried for aging, so that males might also be aged, as well as females past the reproductive period. The best structure is the cylinder of wax which fills the outer canal of the ear leading to the eardrum. This enlarges as the whale grows, and the growth is marked by banding very similar in appearance to tree rings. Slicing the plug, the rings can be exposed and counted. There are produced but two rings a year. Toothed whales can be aged by sawing the teeth in half and noting the banding. Baleen also is periodically enlarged, but this is more difficult to check.

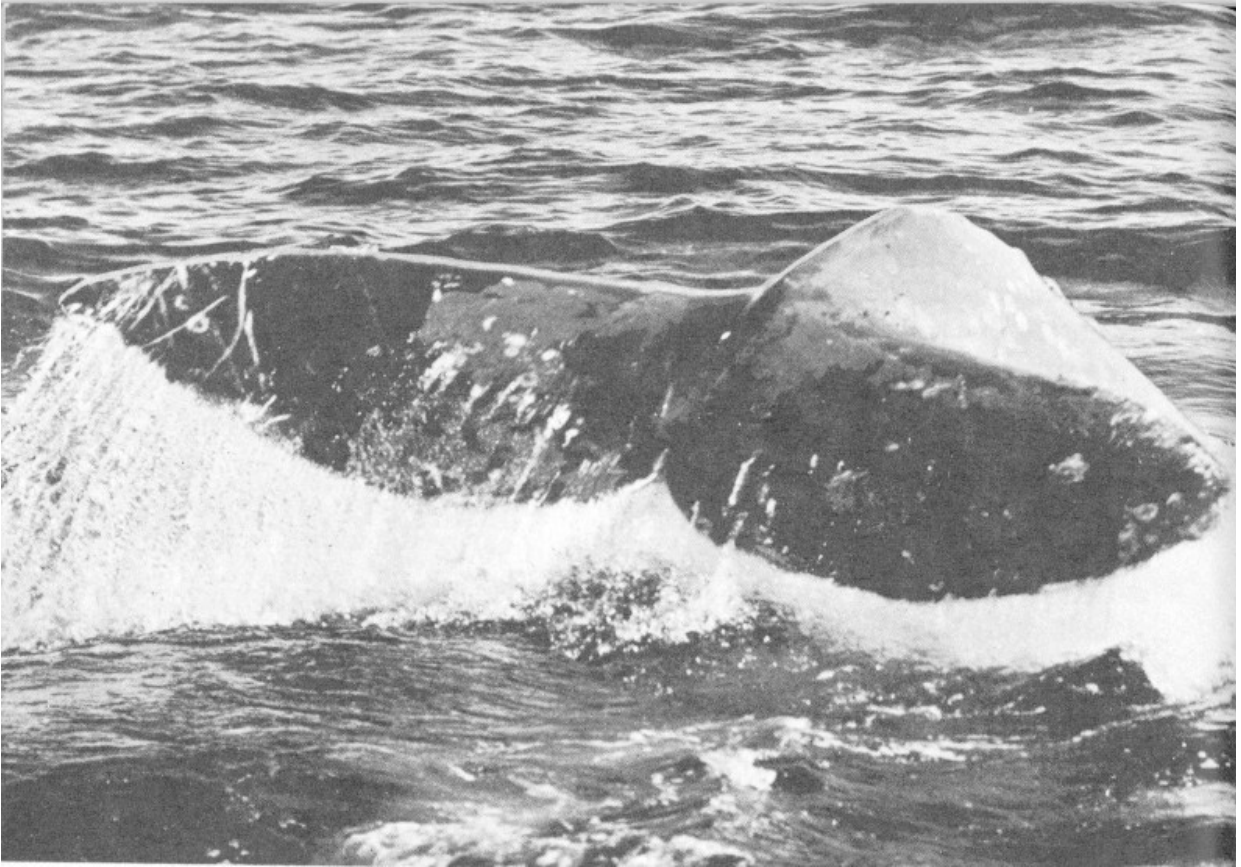
Distribution and Population Rise

The whaling industry spends a few weeks before the whaling season and at the end, hunting whales with a much lighter gun which implants a numbered cylinder deep in the skin of the whale. Each whale so marked is logged as to the species and the latitude and the longitude, and lastly, the date. Should the marked whale be processed in any succeeding season, a second entry is completed giving the date and place of capture. These records are used to determine the migration routes and the degree of intermingling of adjacent populations. The ratio of marked whales to unmarked whales allows the estimation of the probable size of the whale population. The length of time elapsing until a marked whale is recovered is an indication of the intensity of whaling.

Unlike the situation in the fishing industry where the reproductive potential of the fish is more than adequate to repopulate quickly, in the whaling industry the reproductive potential is small. Only one-fourth of the population can contribute replacements each year, and then only a single baby. Of course, each offspring has a reasonable chance of reaching maturity, whereas with fish only a very small number achieve adulthood.

One might expect that whales would be less affected by the fluctuations in environmental conditions, but this is only a surmise. We do not know as yet how to ascertain whether the food of whales is ever inadequate. Certainly the fact that baleen whales do best around the polar seas suggests that climatic fluctuations of the magnitude and duration of those producing the ice ages, may have profound effects on them.

We can only hope that man will learn to exploit whales in such a way as to take only the surplus and leave the main stock, creating thereby a perpetual resource. Man can best profit esthetically and scientifically by leaving the gray whale free of whaling as a control population.



Gray whale making a deep dive. *Photograph by T. J. Walker.*



Gray whale spouting. *Courtesy Scripps Institute of Oceanography.*

Appendix

Scientific names of all North American cetaceans with common names most in use. Maximum size indicated.

Region: (A) Atlantic Ocean, (P) Pacific Ocean, (C) Circumpolar

	<i>Common Name</i>	<i>Size- Region</i>
Order Cetacea	Whales, Dolphins, and Porpoises	
Suborder Odontoceti	Toothed Whales	
Family Ziphiidae	Beaked Whales	
Genus Berardius		
<i>Berardius bairdii</i>	Baird's Beaked Whale	42 ft. (P)
Genus Mesoplodon		
<i>Mesoplodon bidens</i>	Sowerby's Beaked Whale	16 ft. (A)
<i>Mesoplodon densirostris</i>	Blainville's Beaked Whale	15 ft. (A)
<i>Mesoplodon europaeus</i>	Gervais' Beaked Whale	16 ft. (A)
<i>Mesoplodon mirus</i>	True's Beaked Whale	17 ft. (A)
<i>Mesoplodon stejnegeri</i>	Stejneger's Beaked Whale	20 ft. (P)
Genus Ziphius		
<i>Ziphius Cavirostris</i>	Cuvier's Beaked Whale	28 ft. (A- P)
Genus Hyperoodon		
<i>Hyperoodon ampullatus</i>	Flat-headed Bottlenosed Whale	30 ft. (A)
Family Physeteridae	Sperm Whales	
Genus Physeter		
<i>Physeter catodon</i>	Sperm Whale	60 ft. (A- P)

Family Kogiidae	Pigmy Sperm Whales		
Genus Kogia			
<i>Kogia breviceps</i>	Pigmy Sperm Whale	13 ft.	(A-P)
Family Vonodontidae	White Whale and Narwhal		
Genus Delphinapterus			
<i>Delphinapterus leucas</i>	White Whale (Beluga)	18 ft.	(C)
Genus Monodon			
<i>Monodon monoceros</i>	Narwhal	12 ft.	(C)
Family Delphinidae	Dolphins and Porpoises		
Genus Stenella			
<i>Stenella attenuata</i>	Slender-beaked Porpoise	6 ft.	(P)
<i>Stenella frontalis</i>	Cuvier's Porpoise	6 ft.	(A)
<i>Stenella graffmani</i>	Graffman's Porpoise	8 ft.	(P)
<i>Stenella longirostris</i>	Long-beaked Porpoise	7 ft.	(P)
<i>Stenella microps</i>	Small-headed Porpoise	8 ft.	(P)
<i>Stenella plagiodon</i>	Spotted Porpoise	7 ft.	(A)
<i>Stenella styx</i>	Gray's Porpoise	8 ft.	(A-P)
Genus Steno			
<i>Steno bredanensis</i>	Rough-toothed Porpoise	8 ft.	(A-P)
Genus Delphinus			
<i>Delphinus delphis</i>	Atlantic Dolphin	8 ft.	(A)
<i>Delphinus bairdii</i>	Pacific Dolphin	7 ft.	(P)
Genus Tursiops			
<i>Tursiops truncatus</i>	Atlantic Bottle-nosed Dolphin	12 ft.	(A)
<i>Tursiops gillii</i>	Gill's Bottle-nosed Dolphin	12 ft.	(P)
<i>Tursiops nuuanu</i>	Pacific Bottle-nosed Dolphin	7 ft.	(P)
Genus Lissodelphis			

	<i>Lissodelphis borealis</i>	Right-whale Dolphin	8 ft.	(P)
	Genus Lagenorhynchus			
	<i>Lagenorhynchus</i>	White-Beaked Dolphin	10 ft.	(A)
<i>albirostris</i>				
	<i>Lagenorhynchus acutus</i>	Atlantic White-sided Dolphin	9 ft.	(A)
	<i>Lagenorhynchus</i>	Pacific White-sided Dolphin	7 ft.	(P)
<i>obliquidens</i>				
	<i>Lagenorhynchus</i>	Gray's White-sided Dolphin	9 ft.	(P)
<i>thicola</i>				
	Genus Grampus			
	<i>Grampus orca</i>	Atlantic Killer Whale	30 ft.	(A)
	<i>Grampus rectipinna</i>	Pacific Killer Whale	30 ft.	(P)
	Genus Grampidelphis			
	<i>Grampidelphis Griseus</i>	Grampus or Risso's Dolphin	13 ft.	(A-P)
	Genus Pseudorca			
	<i>Pseudorca crassidens</i>	False Killer Whale	18 ft.	(A-P)
	Genus Globicephala			
	<i>Globicephala melaena</i>	Common Blackfish or (Pilot Whale)	28 ft.	(A)
	<i>Globicephala</i>	Short-finned Blackfish	20 ft.	(A)
<i>macrorhyncha</i>				
	<i>Globicephala scammonii</i>	Pacific Blackfish	16 ft.	(P)
	Genus Feresa			
	<i>Feresa occulta</i>	Slender Blackfish	8 ft.	(P)
	Genus Phocoena			
	<i>Phocoena phocoena</i>	Atlantic Harbor Porpoise	6 ft.	(A)
	<i>Phocoena vomerina</i>	Pacific Harbor Porpoise	6 ft.	(A)
	Genus Phocoenoides			
	<i>Phocoenoides dalli</i>	Dall's Porpoise	7 ft.	(P)
	Suborder Mysticeti	Baleen Whales		

Family Eschrichtidae	Gray Whale		
Genus Eschrichtius			
<i>Eschrichtius glaucus</i>	California Gray Whale	50 ft.	(P)
Family Balaenopteridae	Fin-backed Whales		
Genus Balaenoptera			
<i>Balaenoptera physalus</i>	Fin-backed Whale	81 ft.	(A-P)
<i>Balaenoptera borealis</i>	Rorqual (Sei Whale)	60 ft.	(A-P)
<i>Balaenoptera acutorostrata</i>	Little Piked Whale	33 ft.	(A-P)
Genus Sibbaldus			
<i>Sibbaldus musculus</i>	Blue (Sulphur-bottom) Whale	100 ft.	(A-P)
Genus Megaptera			
<i>Megaptera novaeangliae</i>	Hump-backed Whale	50 ft.	(A-P)
Family Balaenidae	Right and Bowhead Whales		
Genus Eubalaena			
<i>Eubalaena glacialis</i>	Atlantic Right Whale	70 ft.	(A)
<i>Eubalaena sieboldii</i>	Pacific Right Whale	65 ft.	(P)
Genus Balaena			
<i>Balaena mysticetus</i>	Bowhead Whale	65 ft.	(C)

Transcriber's Notes

- Retained publication information from the printed edition: this eBook is public-domain in the country of publication.
- Silently corrected a few palpable typos.
- In the text versions only, text in italics is delimited by _underscores_.

*** END OF THE PROJECT GUTENBERG EBOOK WHALE PRIMER,
WITH SPECIAL ATTENTION TO THE CALIFORNIA GRAY WHALE

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